

Graph Neural Networks and Graph Foundation Models for Scientific Discovery: A Survey

Graphs are a natural language for scientific structure: atoms form molecular graphs, crystals form periodic lattices, proteins form residue and interaction graphs, simulations use meshes and particles, and scientific knowledge is linked through entities, relations, and evidence. Graph neural networks and graph foundation models therefore provide a unified computational substrate for scientific discovery. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around message passing, graph transformers, geometric equivariance, molecular and materials modeling, protein and biomedical graphs, physical simulation, graph pretraining, scientific knowledge graphs, and benchmarks. The survey argues that the next generation of graph foundation models for science must combine structural priors, scale, uncertainty, domain constraints, and interpretable discovery workflows.

Keywords:

graph neural networks; graph foundation models; scientific discovery; molecular graphs; materials discovery; protein graphs; geometric deep learning; knowledge graphs

1 Introduction

Scientific data are often relational. Molecules are graphs of atoms and bonds, crystals are periodic graphs of atoms and symmetries, proteins contain residue contacts and interaction networks, physical simulations evolve particles or meshes, and scientific literature forms knowledge graphs linking hypotheses, methods, datasets, and discoveries. Graph neural networks (GNNs) exploit this structure by learning representations that respect nodes, edges, neighborhoods, and symmetries [21, 29, 73, 151, 161, 174, 216, 232].

Graph foundation models extend this idea from task-specific graph predictors to large pretrained systems that can adapt across molecules, materials, proteins, physical systems, and knowledge graphs. The promise is scientific transfer: a model that learns reusable structural and geometric priors from large corpora may reduce data requirements for new experiments, accelerate screening, and support hypothesis generation [13, 21, 65, 89, 149, 161, 174, 232]. The risk is that scale alone may hide weak physical valid-

ity, poor uncertainty, or benchmark leakage.

This survey reviews graph neural networks and graph foundation models for scientific discovery. We focus on architectures, scientific domains, pretraining, evaluation, interpretability, and deployment in discovery workflows.

2 Background

2.1 Graph Neural Networks

GNNs learn node, edge, graph, or geometric representations through local aggregation and nonlinear transformations. Message passing neural networks provide the canonical formulation; graph transformers and higher-order methods expand receptive fields and expressivity; geometric models encode equivariance to rotations, translations, and permutations. These architectural choices matter in science because molecules, materials, and physical systems obey symmetries and conservation laws [21, 30, 70, 94, 151, 174, 216, 232].

2.2 Scientific Discovery as Graph Learning

Scientific discovery requires more than predictive accuracy. A model may propose molecules, rank materials, predict protein interactions, simulate dynamics, or connect literature evidence. Good graph models must therefore balance representation learning with domain constraints, uncertainty, causal reasoning, and experimental validation. A discovery model is useful when it narrows the search space, exposes explanations, and remains robust under distribution shift.

2.3 Graph Foundation Models

Graph foundation models seek broad transfer through large-scale graph pretraining, self-supervised objectives, masked attribute prediction, contrastive learning, generative modeling, instruction tuning, or multimodal alignment. In science, these models must handle heterogeneous graph sizes, continuous geometry, periodic boundary conditions, sparse labels, and multiple measurement modalities [13, 21, 65, 89, 149, 161, 174, 232].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with graph neural networks, graph foundation models, graph transformers, geometric deep learning, equivariant GNNs, message passing neural networks, molecular graphs, drug discovery, property prediction, quantum chemistry, materials, crystals, catalysis, proteins, biomedical graphs, physical simulation, neural operators, mesh graph networks, scientific knowledge graphs, OGB, Open Catalyst Project, MatBench, PCQM4M, QM9, and MoleculeNet. Records were retained when title or abstract evidence connected graph learning with scientific discovery, chemistry, materials, biology, physics, simulation, scientific knowledge graphs, pretraining, or benchmark evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the field along six axes. The first is *graph type*: molecular graphs, periodic crystal

graphs, protein contact graphs, biomedical interaction graphs, meshes, particles, scientific knowledge graphs, or heterogeneous experimental graphs. The second is *architecture*: message passing, graph transformer, equivariant network, neural operator, generative graph model, or hybrid geometric model. The third is *learning signal*: supervised property prediction, self-supervision, contrastive learning, masked modeling, generative design, or simulation loss. The fourth is *scientific task*: screening, design, dynamics, reaction prediction, structure prediction, hypothesis generation, or literature reasoning. The fifth is *validation*: benchmark metrics, uncertainty, physical consistency, ablation, and experimental verification. The sixth is *foundation-model behavior*: transfer, scaling, adaptation, multimodal alignment, and prompt or instruction interfaces.

3.3 Architectures and Geometric Priors

Message passing gives GNNs a natural inductive bias for local interactions. Scientific domains often require stronger priors: rotational equivariance for 3D molecular and materials modeling, periodicity for crystals, conservation for physical simulation, and long-range attention for large molecules or knowledge graphs. Graph transformers and equivariant architectures therefore occupy a central place in modern scientific graph learning [21, 30, 70, 94, 151, 174, 216, 232].

3.4 Molecular Graphs and Drug Discovery

Molecular GNNs support property prediction, virtual screening, reaction modeling, generative design, and protein-ligand interaction modeling. The graph abstraction is powerful because atoms and bonds define a chemically meaningful structure, but scientific usefulness depends on stereochemistry, conformers, quantum effects, synthesis constraints, and experimental noise. Graph foundation models may improve transfer across assays and molecular tasks, but they must avoid learning dataset shortcuts [21, 94, 138, 151, 161, 174, 216, 232].

3.5 Materials, Crystals, and Catalysis

Materials discovery uses graph models for formation energy, band gaps, adsorption, crystal stability, catalyst screening, and interatomic potentials. Periodic structure and symmetry are essential: a material is not only a finite graph but a repeating geometric object with physical constraints. Graph mod-

els for materials increasingly combine equivariance, periodic neighbor construction, and large-scale pre-training over computed or experimental databases [21, 29, 65, 73, 94, 138, 161, 174].

3.6 Proteins, Biology, and Biomedical Graphs

Biological discovery uses graphs at multiple scales: residue contacts, protein-ligand complexes, protein-protein interactions, gene networks, disease pathways, and biomedical knowledge graphs. These settings combine structural, sequence, text, and experimental signals. GNNs can integrate such heterogeneous evidence, while graph foundation models may support transfer across biological tasks and limited-label regimes [12, 39, 101, 124, 131, 151, 261, 270].

3.7 Physical Simulation and Scientific Machine Learning

Graph networks model particles, meshes, fluids, solids, and PDE systems by treating local interactions as edges. This approach aligns with numerical simulation, where locality and geometry matter. Neural operators and mesh graph networks extend graph learning to continuous fields and time-dependent systems. The challenge is preserving stability, conservation properties, and extrapolation under new boundary conditions [18, 20, 21, 30, 71, 73, 125, 256].

3.8 Scientific Knowledge Graphs and Discovery Workflows

Knowledge graphs encode relations among entities, publications, methods, materials, molecules, genes, and experimental observations. Scientific discovery workflows can use graph reasoning to retrieve evidence, propose hypotheses, identify missing links, and connect mechanistic explanations to predictions. Graph foundation models for knowledge graphs must handle provenance, uncertainty, temporal change, and heterogeneous relation semantics [14, 73, 88, 124–126, 222, 236].

3.9 Benchmarks and Evaluation

Benchmarks make progress measurable but can also narrow the field. Scientific graph benchmarks should report data splits, scaffold or time-based generalization, out-of-distribution behavior, uncertainty, physical consistency, and experimental validation where possible. Leaderboard gains are less meaningful when the proposed molecule is unsynthesizable, the

material violates constraints, or a physics surrogate is unstable [21, 94, 138, 151, 161, 174, 216, 232].

3.10 Representative Literature Map

The following map cites the full retained corpus by theme, making the coverage explicit across architectures, scientific domains, graph foundation models, and evaluation.

Gnn architectures and geometric graph learning

The gnn architectures and geometric graph learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [21, 30, 65, 70, 149, 155, 174, 216, 270]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [14, 20, 39, 71, 83, 126, 144, 193, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 51, 132, 167, 178, 233, 240, 261, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [19, 40, 41, 56, 76, 204, 243, 245, 249]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [57, 115, 118, 130, 171, 212, 289, 291, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 97, 134, 142, 160, 200, 207, 263, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [33, 38, 63, 98, 121, 145, 241, 262, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 52, 100, 107, 108, 148, 176, 255, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 17, 42, 136, 173, 188, 217, 226, 280]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [77, 82, 93, 95, 209, 218–220, 265]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 24, 48, 105, 109, 116, 133, 137, 180]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 92, 114, 117, 120, 185, 197, 231, 292]. Additional retained studies broaden the cluster with alternative architectures, training

signals, data sources, and deployment assumptions [61, 62, 99, 156, 202, 228, 235, 244, 247]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [59, 210, 227, 259].

Molecular representation learning and drug discovery The molecular representation learning and drug discovery cluster contains work that develops methods, systems, datasets, or analyses central to this survey [86, 94, 138, 151, 161, 232, 256, 269, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [12, 13, 89, 96, 131, 183, 222, 239, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [36, 66, 74, 75, 101, 103, 124, 154, 224]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [11, 80, 181, 198, 203, 225, 251, 271, 281]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [64, 79, 91, 140, 164, 166, 253, 267, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 25, 34, 163, 192, 205, 260, 275, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 135, 179, 189, 190, 229, 258, 276, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [35, 102, 159, 162, 213, 214, 223, 257, 266]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [23, 58, 90, 139, 196, 211, 237, 254, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [43, 81, 87, 150, 152, 165, 184, 215, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [44, 60, 182, 201, 208, 230, 246, 268, 277]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [32, 49, 106, 143, 158, 234, 252, 274, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [2, 67, 104, 122, 127, 242, 283].

Materials, crystals, and catalysis The materials, crystals, and catalysis cluster contains work that develops methods, systems, datasets, or analyses central to this survey [29, 47, 50, 53, 69, 153, 194, 195, 286]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [45, 46, 54, 78, 146, 168, 172, 186, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 27, 31, 68, 84, 123, 187, 238, 264]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 37, 111, 141, 175, 199, 221, 272, 273].

Proteins, biology, and biomedical graphs The proteins, biology, and biomedical graphs cluster contains work that develops methods, systems, datasets, or analyses central to this survey [7, 16, 28, 85, 147, 157, 206, 248, 298]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [55, 170].

Physical simulation and scientific machine learning The physical simulation and scientific machine learning cluster contains work that develops methods, systems, datasets, or analyses central to this survey [8, 73, 110].

Graph foundation models and pretraining The graph foundation models and pretraining cluster contains work that develops methods, systems, datasets, or analyses central to this survey [177].

Scientific knowledge graphs and discovery workflows The scientific knowledge graphs and discovery workflows cluster contains work that develops methods, systems, datasets, or analyses central to this survey [125].

Datasets, benchmarks, and evaluation The datasets, benchmarks, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [72, 88, 112, 113, 119, 169, 191, 236, 250]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [3, 128, 129].

4 Challenges and Prospects

4.1 Scaling Without Losing Scientific Validity

Graph foundation models can scale across tasks and datasets, but science imposes constraints that generic pretraining may ignore. Future models should combine data scale with physical priors, uncertainty, conservation laws, and domain-specific validation.

4.2 Out-of-Distribution Generalization

Scientific discovery is often about extrapolation: new molecules, new materials, new proteins, or new regimes. Evaluation should emphasize scaffold splits, temporal splits, composition shifts, new boundary conditions, and prospective validation rather than only random splits.

4.3 Uncertainty and Experimental Decision Making

A model for discovery should know when it is uncertain. Calibration, ensembles, Bayesian graph learning, conformal prediction, and active learning can help prioritize experiments and avoid overconfident false discoveries.

4.4 Multimodal Scientific Graphs

Future graph foundation models will combine structure with spectra, text, images, simulations, assays, and laboratory metadata. The central challenge is aligning these modalities without erasing the graph and geometric structure that makes scientific reasoning possible.

4.5 From Prediction to Closed-Loop Discovery

The long-term prospect is a closed-loop workflow in which graph models propose candidates, estimate uncertainty, retrieve evidence, suggest experiments, update from results, and maintain provenance. This requires integration with laboratory automation, literature systems, and human scientific judgment.

5 Conclusion

Graph neural networks and graph foundation models are central tools for scientific discovery because they encode the relational structure of molecules, materials, proteins, physical systems, and scientific knowledge. This survey reviewed architectures, molecular and materials modeling, biological graphs, physi-

cal simulation, graph pretraining, knowledge graphs, and evaluation. The central lesson is that graph foundation models for science must be judged not only by benchmark accuracy but by validity, uncertainty, transfer, interpretability, and usefulness in real discovery loops.

References

- [1] Frank M. Abel, Jaehyung Lee, Charles R. Campbell, and Kamal Choudhary. RamanGPT: Bidirectional Mapping Between Crystal Structures and Raman Spectra with Graph Neural Networks and Generative Transformers. *arXiv preprint arXiv:2606.03764*, 2026. URL <https://arxiv.org/abs/2606.03764>.
- [2] Stefan Abi-Karam, Yuqi He, Rishov Sarkar, Lakshmi Sathidevi, Zihang Qiao, and Cong Hao. GenGNN: A Generic FPGA Framework for Graph Neural Network Acceleration. *arXiv preprint arXiv:2201.08475*, 2022. URL <https://arxiv.org/abs/2201.08475>.
- [3] Jakub Adamczyk. Application of Graph Neural Networks and graph descriptors for graph classification. *arXiv preprint arXiv:2211.03666*, 2022. URL <https://arxiv.org/abs/2211.03666>.
- [4] Matthew Adrian, Yunsie Chung, Kevin Boyd, Saeed Paliwal, Srimukh Prasad Veccham, and Alan C. Cheng. Multitask finetuning and acceleration of chemical pretrained models for small molecule drug property prediction. *arXiv preprint arXiv:2510.12719*, 2025. URL <https://arxiv.org/abs/2510.12719>.
- [5] Salma Afifi, Febin Sunny, Amin Shafiee, Mahdi Nikdast, and Sudeep Pasricha. GHOST: A Graph Neural Network Accelerator using Silicon Photonics. *arXiv preprint arXiv:2307.01782*, 2023. URL <https://arxiv.org/abs/2307.01782>.
- [6] Mansoor Ahmed, Sujin Lee, Umar Khayaz, and Murray Patterson. EvoStruct: Bridging Evolutionary and Structural Priors for Antibody CDR Design via Protein Language Model Adaptation. *arXiv preprint arXiv:2605.21485*, 2026. URL <https://arxiv.org/abs/2605.21485>.
- [7] Justin Airas and Bin Zhang. Scaling Graph Neural Networks to Large Proteins. *arXiv preprint arXiv:2410.03921*, 2024. URL <https://arxiv.org/abs/2410.03921>.
- [8] Justin Airas and Bin Zhang. Knowledge Distillation of a Protein Language Model Yields a Foundational Implicit Solvent Model. *arXiv preprint arXiv:2601.05388*, 2026. URL <https://arxiv.org/abs/2601.05388>.

- [9] Lilas Alrahis, Johann Knechtel, and Ozgur Sinanoglu. Graph Neural Networks: A Powerful and Versatile Tool for Advancing Design, Reliability, and Security of ICs. *arXiv preprint arXiv:2211.16495*, 2022. URL <https://arxiv.org/abs/2211.16495>.
- [10] Gopal Anantharaman. A Topological Characterization of Graph Neural Networks via Stochastic Block Model Embeddings on the n-Sphere. *arXiv preprint arXiv:2606.07598*, 2026. URL <https://arxiv.org/abs/2606.07598>.
- [11] Daniele Angioletti, Stefano Raniolo, and Vittorio Limongelli. HEroBM: a deep equivariant graph neural network for universal backmapping from coarse-grained to all-atom representations. *arXiv preprint arXiv:2404.16911*, 2024. URL <https://arxiv.org/abs/2404.16911>.
- [12] Ramin Ansari and Amirata Ghorbani. Accurate Prediction of Free Solvation Energy of Organic Molecules via Graph Attention Network and Message Passing Neural Network from Pairwise Atomistic Interactions. *arXiv preprint arXiv:2105.02048*, 2021. URL <https://arxiv.org/abs/2105.02048>.
- [13] Boshra Ariguib, Mathias Niepert, and Andrei Manolache. Learning the Neighborhood: Contrast-Free Multimodal Self-Supervised Molecular Graph Pretraining. *arXiv preprint arXiv:2509.22468*, 2025. URL <https://arxiv.org/abs/2509.22468>.
- [14] Abderaouf Bahi. Graph Neural Networks Applications Across Domains: All Insights You Need. *arXiv preprint arXiv:2606.27202*, 2026. URL <https://arxiv.org/abs/2606.27202>.
- [15] Maria Bankestad, Keven M. Dorst, Goran Widmalm, and Jerk Ronnols. Carbohydrate NMR chemical shift predictions using E(3) equivariant graph neural networks. *arXiv preprint arXiv:2311.12657*, 2023. URL <https://arxiv.org/abs/2311.12657>.
- [16] Jose Manuel Barraza-Chavez, Rana A. Barghout, Ricardo Almada-Monter, Adrian Jinich, Radhakrishnan Mahadevan, and Benjamin Sanchez-Lengeling. Graph Data Modeling: Molecules, Proteins, & Chemical Processes. *arXiv preprint arXiv:2508.19356*, 2025. URL <https://arxiv.org/abs/2508.19356>.
- [17] Ilyes Batatia, David Peter Kovacs, Gregor N. C. Simm, Christoph Ortner, and Gabor Csanyi. MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields. *arXiv preprint arXiv:2206.07697*, 2022. URL <https://arxiv.org/abs/2206.07697>.
- [18] Simon Batzner, Albert Musaelian, Lixin Sun, Mario Geiger, Jonathan P. Mailoa, Mordechai Kornbluth, Nicola Molinari, Tess E. Smidt, and Boris Kozinsky. E(3)-Equivariant Graph Neural Networks for Data-Efficient and Accurate Interatomic Potentials. *arXiv preprint arXiv:2101.03164*, 2021. URL <https://arxiv.org/abs/2101.03164>.
- [19] Satadeep Bhattacharjee. SR-CGCNN: Shared Recurrent Convolution in Crystal Graph Neural Networks for Materials Property Prediction. *arXiv preprint arXiv:2605.01304*, 2026. URL <https://arxiv.org/abs/2605.01304>.
- [20] Shuyu Bi, Zhede Zhao, Qiangchao Sun, Tao Hu, Xionggang Lu, and Hongwei Cheng. Universal and efficient graph neural networks with dynamic attention for machine learning interatomic potentials. *arXiv preprint arXiv:2603.22810*, 2026. URL <https://arxiv.org/abs/2603.22810>.
- [21] Vaibhav Bihani, Utkarsh Pratiush, Sajid Mannan, Tao Du, Zhimin Chen, Santiago Miret, Matthieu Micoulaut, Morten M Smedskjaer, Sayan Ranu, and N M Anoop Krishnan. EGraFFBench: Evaluation of Equivariant Graph Neural Network Force Fields for Atomistic Simulations. *arXiv preprint arXiv:2310.02428*, 2023. URL <https://arxiv.org/abs/2310.02428>.
- [22] Pietro Bongini, Monica Bianchini, and Franco Scarselli. Molecular graph generation with Graph Neural Networks. *arXiv preprint arXiv:2012.07397*, 2020. URL <https://arxiv.org/abs/2012.07397>.
- [23] Niklas Breustedt, Paolo Climaco, Jochen Garcke, Jan Hamaekers, Gitta Kutyniok, Dirk A. Lorenz, Rick Oerder, and Chirag Varun Shukla. On the Interplay of Subset Selection and Informed Graph Neural Networks. *arXiv preprint arXiv:2306.10066*, 2023. URL <https://arxiv.org/abs/2306.10066>.
- [24] Fabio Broccatelli, Richard Trager, Michael Reutlinger, George Karypis, and Mufei Li. Benchmarking Accuracy and Generalizability of Four Graph Neural Networks Using Large In Vitro ADME Datasets from Different Chemical Spaces. *arXiv preprint arXiv:2111.13964*, 2021. URL <https://arxiv.org/abs/2111.13964>.
- [25] Jonas Busk, Peter Bjørn Jrgensen, Arghya Bhowmik, Mikkel N. Schmidt, Ole Winther, and Tejs Vegge. Calibrated Uncertainty for Molecular Property Prediction using Ensembles of Message Passing Neural Networks. *arXiv preprint arXiv:2107.06068*, 2021. URL <https://arxiv.org/abs/2107.06068>.
- [26] Yang Cao, Zhao Song, Jiahao Zhang, and Jiale Zhao. Fundamental Limits of Crystalline Equivariant Graph Neural Networks: A Circuit Complexity Perspective. *arXiv preprint arXiv:2510.05494*, 2025. URL <https://arxiv.org/abs/2510.05494>.
- [27] Alvaro Carbonero, Alexandre Duval, Victor Schmidt, Santiago Miret, Alex Hernandez-Garcia, Yoshua Bengio, and David Rolnick. On the

- importance of catalyst-adsorbate 3D interactions for relaxed energy predictions. *arXiv preprint arXiv:2310.06682*, 2023. URL <https://arxiv.org/abs/2310.06682>.
- [28] Francesco Ceccarelli, Lorenzo Giusti, Sean B. Holden, and Pietro Lio. Neural Embeddings for Protein Graphs. *arXiv preprint arXiv:2306.04667*, 2023. URL <https://arxiv.org/abs/2306.04667>.
- [29] Achuth Chandrasekhar, Janghoon Ock, and Amir Barati Farimani. Catalyst-Agent: Autonomous heterogeneous catalyst screening with an LLM Agent. *arXiv preprint arXiv:2603.01311*, 2026. URL <https://arxiv.org/abs/2603.01311>.
- [30] Jian Chang and Shuze Zhu. MGNN: Moment Graph Neural Network for Universal Molecular Potentials. *arXiv preprint arXiv:2409.15800*, 2024. URL <https://arxiv.org/abs/2409.15800>.
- [31] Lowik Chanussot, Abhishek Das, Siddharth Goyal, Thibaut Lavril, Muhammed Shuaibi, Morgane Riviere, Kevin Tran, Javier Heras-Domingo, Caleb Ho, Weihua Hu, Aini Palizhati, Anuroop Sriram, Brandon Wood, Junwoong Yoon, Devi Parikh, C. Lawrence Zitnick, and Zachary Ulissi. The Open Catalyst 2020 (OC20) Dataset and Community Challenges. *arXiv preprint arXiv:2010.09990*, 2020. URL <https://arxiv.org/abs/2010.09990>.
- [32] Benson Chen, Regina Barzilay, and Tommi Jaakkola. Path-Augmented Graph Transformer Network. *arXiv preprint arXiv:1905.12712*, 2019. URL <https://arxiv.org/abs/1905.12712>.
- [33] Chen Chen, Enze Xu, Defu Yang, Chenggang Yan, Tao Wei, Hanning Chen, Yong Wei, and Minghan Chen. Chemical Environment Adaptive Learning for Optical Band Gap Prediction of Doped Graphitic Carbon Nitride Nanosheets. *arXiv preprint arXiv:2302.09539*, 2023. URL <https://arxiv.org/abs/2302.09539>.
- [34] Jiahao Chen, Yurou Liu, Jiangmeng Li, Bing Su, and Jirong Wen. Atomic and Subgraph-aware Bilateral Aggregation for Molecular Representation Learning. *arXiv preprint arXiv:2305.12618*, 2023. URL <https://arxiv.org/abs/2305.12618>.
- [35] Mengjie Chen, Ming Zhang, and Cunquan Qu. Towards Interpretable Drug-Drug Interaction Prediction: A Graph-Based Approach with Molecular and Network-Level Explanations. *arXiv preprint arXiv:2507.09173*, 2025. URL <https://arxiv.org/abs/2507.09173>.
- [36] Pengfei Chen, Weiwen Liu, Chang-Yu Hsieh, Guangyong Chen, and Shengyu Zhang. Utilizing Edge Features in Graph Neural Networks via Variational Information Maximization. *arXiv preprint arXiv:1906.05488*, 2019. URL <https://arxiv.org/abs/1906.05488>.
- [37] Wenjie Chen, Zichang Lin, Xinxin Zhang, Hao Zhou, and Yuegang Zhang. AI-Driven Accelerated Discovery of Intercalation-type Cathode Materials for Magnesium Batteries. *arXiv preprint arXiv:2412.11032*, 2024. URL <https://arxiv.org/abs/2412.11032>.
- [38] Zhengdao Chen, Lei Chen, Soledad Villar, and Joan Bruna. Can Graph Neural Networks Count Substructures? *arXiv preprint arXiv:2002.04025*, 2020. URL <https://arxiv.org/abs/2002.04025>.
- [39] Bryan Cheng and Austin Jin. Structure-Aware Prediction of PROTAC-Mediated Protein Degradability via Graph Neural Networks. *arXiv preprint arXiv:2606.04021*, 2026. URL <https://arxiv.org/abs/2606.04021>.
- [40] Seungyeon Choi, Sangmin Seo, and Sanghyun Park. SPIN: SE(3)-Invariant Physics Informed Network for Binding Affinity Prediction. *arXiv preprint arXiv:2407.11057*, 2024. URL <https://arxiv.org/abs/2407.11057>.
- [41] Kamal Choudhary and Brian DeCost. Atomistic Line Graph Neural Network for Improved Materials Property Predictions. *arXiv preprint arXiv:2106.01829*, 2021. URL <https://arxiv.org/abs/2106.01829>.
- [42] Hejie Cui, Wei Dai, Yanqiao Zhu, Xuan Kan, Antonio Aodong Chen Gu, Joshua Lukemire, Liang Zhan, Lifang He, Ying Guo, and Carl Yang. BrainGB: A Benchmark for Brain Network Analysis with Graph Neural Networks. *arXiv preprint arXiv:2204.07054*, 2022. URL <https://arxiv.org/abs/2204.07054>.
- [43] Markus Dablander. Investigating Graph Neural Networks and Classical Feature-Extraction Techniques in Activity-Cliff and Molecular Property Prediction. *arXiv preprint arXiv:2411.13688*, 2024. URL <https://arxiv.org/abs/2411.13688>.
- [44] Hanjun Dai, Chengtao Li, Connor W. Coley, Bo Dai, and Le Song. Retrosynthesis Prediction with Conditional Graph Logic Network. *arXiv preprint arXiv:2001.01408*, 2020. URL <https://arxiv.org/abs/2001.01408>.
- [45] Kishalay Das, Bidisha Samanta, Pawan Goyal, Seung-Cheol Lee, Satadeep Bhattacharjee, and Niloy Ganguly. CrysGNN : Distilling pre-trained knowledge to enhance property prediction for crystalline materials. *arXiv preprint arXiv:2301.05852*, 2023. URL <https://arxiv.org/abs/2301.05852>.
- [46] Auguste de Lambilly, Vladimir Baturin, David Portehault, Guillaume Lambard, Nataliya Sokolovska, Florence d’Alche Buc, and Jean-Claude Crivello. Finetuning-Free Diffusion Model with Adaptive Constraint Guidance for Inorganic Crystal Structure Generation.

- arXiv preprint arXiv:2604.13354*, 2026. URL <https://arxiv.org/abs/2604.13354>.
- [47] Bowen Deng. Catalysis distillation neural network for the few shot open catalyst challenge. *arXiv preprint arXiv:2305.19545*, 2023. URL <https://arxiv.org/abs/2305.19545>.
- [48] Juergen Dietrich. From Detection to Mechanism: Cross-Attention Graph Neural Networks Enable Drug-Drug Interaction Type Prediction An Ablation Study with Acetylsalicylic Acid Validation. *arXiv preprint arXiv:2605.27861*, 2026. URL <https://arxiv.org/abs/2605.27861>.
- [49] Kien Do, Truyen Tran, and Svetha Venkatesh. Graph Transformation Policy Network for Chemical Reaction Prediction. *arXiv preprint arXiv:1812.09441*, 2018. URL <https://arxiv.org/abs/1812.09441>.
- [50] Hongwei Du and Hong Wang. Unveiling the critical factors in crystal structure graph representation: a comparative analysis using streamlined MLPsets frameworks. *arXiv preprint arXiv:2509.05712*, 2025. URL <https://arxiv.org/abs/2509.05712>.
- [51] Hongwei Du, Jiamin Wang, Jian Hui, Lanting Zhang, and Hong Wang. DenseGNN: universal and scalable deeper graph neural networks for high-performance property prediction in crystals and molecules. *arXiv preprint arXiv:2501.03278*, 2025. URL <https://arxiv.org/abs/2501.03278>.
- [52] Weitao Du, He Zhang, Yuanqi Du, Qi Meng, Wei Chen, Bin Shao, and Tie-Yan Liu. SE(3) Equivariant Graph Neural Networks with Complete Local Frames. *arXiv preprint arXiv:2110.14811*, 2021. URL <https://arxiv.org/abs/2110.14811>.
- [53] Zijian Du, Luozhijie Jin, Le Shu, Yan Cen, Yuanfeng Xu, Yongfeng Mei, and Hao Zhang. CT-GNN: Crystal Transformer Graph Neural Network for Crystal Material Property Prediction. *arXiv preprint arXiv:2405.11502*, 2024. URL <https://arxiv.org/abs/2405.11502>.
- [54] Alexander Dunn, Qi Wang, Alex Ganose, Daniel Dopp, and Anubhav Jain. Benchmarking Materials Property Prediction Methods: The Matbench Test Set and Automatminer Reference Algorithm. *arXiv preprint arXiv:2005.00707*, 2020. URL <https://arxiv.org/abs/2005.00707>.
- [55] Ian Dunn and David Ryan Koes. Accelerating Inference in Molecular Diffusion Models with Latent Representations of Protein Structure. *arXiv preprint arXiv:2311.13466*, 2023. URL <https://arxiv.org/abs/2311.13466>.
- [56] Vijay Prakash Dwivedi, Chaitanya K. Joshi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Benchmarking Graph Neural Networks. *arXiv preprint arXiv:2003.00982*, 2020. URL <https://arxiv.org/abs/2003.00982>.
- [57] Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Graph Neural Networks with Learnable Structural and Positional Representations. *arXiv preprint arXiv:2110.07875*, 2021. URL <https://arxiv.org/abs/2110.07875>.
- [58] Anatol Ehrlich, Lorenz Kummer, Vojtech Voracek, Franka Bause, and Nils M. Kriege. XIMP: Cross Graph Inter-Message Passing for Molecular Property Prediction. *arXiv preprint arXiv:2601.19037*, 2026. URL <https://arxiv.org/abs/2601.19037>.
- [59] Moshe Eliasof, Eldad Haber, and Eran Treister. Graph Neural Reaction Diffusion Models. *arXiv preprint arXiv:2406.10871*, 2024. URL <https://arxiv.org/abs/2406.10871>.
- [60] Yin Fang, Qiang Zhang, Haihong Yang, Xiang Zhuang, Shumin Deng, Wen Zhang, Ming Qin, Zhuo Chen, Xiaohui Fan, and Huajun Chen. Molecular Contrastive Learning with Chemical Element Knowledge Graph. *arXiv preprint arXiv:2112.00544*, 2021. URL <https://arxiv.org/abs/2112.00544>.
- [61] Arthur M. Faria, Ignacio F. Grana, and Savvas Varsamopoulos. Inductive Graph Representation Learning with Quantum Graph Neural Networks. *arXiv preprint arXiv:2503.24111*, 2025. URL <https://arxiv.org/abs/2503.24111>.
- [62] Jean Philip Filling, Felix Post, Michael Wand, and Denis Andrienko. Direct Molecular Polarizability Prediction with SO(3) Equivariant Local Frame GNNs. *arXiv preprint arXiv:2511.07087*, 2025. URL <https://arxiv.org/abs/2511.07087>.
- [63] Jean Philip Filling, Daniel Franzen, and Michael Wand. Tensor Channel Equivariant Graph Neural Networks for Molecular Polarizability Prediction. *arXiv preprint arXiv:2605.16891*, 2026. URL <https://arxiv.org/abs/2605.16891>.
- [64] Stefano Fiorini, Giulia M. Bovolenta, Stefano Coniglio, Michele Ciavotta, Pietro Morerio, Michele Parrinello, and Alessio Del Bue. DLGNet: Hyperedge Classification through Directed Line Graphs for Chemical Reactions. *arXiv preprint arXiv:2410.06969*, 2024. URL <https://arxiv.org/abs/2410.06969>.
- [65] Jesun Firoz, Franco Pellegrini, Mario Geiger, Darren Hsu, Jenna A. Bilbrey, Han-Yi Chou, Maximilian Stadler, Markus Hoehnerbach, Tingyu Wang, Dejun Lin, Emine Kucukbenli, Henry W. Sprueill, Ilyes Batatia, Sotiris S. Xantheas, MalSoon Lee, Chris Mundy, Gabor Csanyi, Justin S. Smith, Ponuswamy Sadayappan, and Sutanay Choudhury. Optimizing Data Distribution and Kernel Performance for Efficient Training of Chemistry Founda-

- tion Models: A Case Study with MACE. *arXiv preprint arXiv:2504.10700*, 2025. URL <https://arxiv.org/abs/2504.10700>.
- [66] Klemens Flöge, Srisruthi Udayakumar, Johanna Sommer, Marie Piraud, Stefan Kesselheim, Vincent Fortuin, Stephan Gunneman, Karel J van der Weg, Holger Gohlke, Erinc Merdivan, and Alina Bazarova. OneProt: Towards Multi-Modal Protein Foundation Models. *arXiv preprint arXiv:2411.04863*, 2024. URL <https://arxiv.org/abs/2411.04863>.
- [67] Piotr Gainski, Micha Koziarski, Jacek Tabor, and Marek Smieja. ChiENN: Embracing Molecular Chirality with Graph Neural Networks. *arXiv preprint arXiv:2307.02198*, 2023. URL <https://arxiv.org/abs/2307.02198>.
- [68] Rongzhi Gao, ChiYung Yam, Jianjun Mao, Shuguang Chen, GuanHua Chen, and Ziyang Hu. A foundation machine learning potential with polarizable long-range interactions for materials modelling. *arXiv preprint arXiv:2410.13820*, 2024. URL <https://arxiv.org/abs/2410.13820>.
- [69] Zachary Garipey, ZhiWen Chen, Isaac Tamblyn, Chandra Veer Singh, and Conrad Giresse Tessassi Feugmo. Automatic graph representation algorithm for heterogeneous catalysis. *arXiv preprint arXiv:2306.04742*, 2023. URL <https://arxiv.org/abs/2306.04742>.
- [70] Johannes Gasteiger, Janek Gro, and Stephan Gunnemann. Directional Message Passing for Molecular Graphs. *arXiv preprint arXiv:2003.03123*, 2020. URL <https://arxiv.org/abs/2003.03123>.
- [71] Johannes Gasteiger, Florian Becker, and Stephan Gunnemann. GemNet: Universal Directional Graph Neural Networks for Molecules. *arXiv preprint arXiv:2106.08903*, 2021. URL <https://arxiv.org/abs/2106.08903>.
- [72] Johannes Gasteiger, Muhammed Shuaibi, Anuroop Sriram, Stephan Gunnemann, Zachary Ulissi, C. Lawrence Zitnick, and Abhishek Das. GemNet-OC: Developing Graph Neural Networks for Large and Diverse Molecular Simulation Datasets. *arXiv preprint arXiv:2204.02782*, 2022. URL <https://arxiv.org/abs/2204.02782>.
- [73] Alireza Ghafarollahi and Markus J. Buehler. Rapid and Automated Alloy Design with Graph Neural Network-Powered LLM-Driven Multi-Agent Systems. *arXiv preprint arXiv:2410.13768*, 2024. URL <https://arxiv.org/abs/2410.13768>.
- [74] Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural Message Passing for Quantum Chemistry. *arXiv preprint arXiv:1704.01212*, 2017. URL <https://arxiv.org/abs/1704.01212>.
- [75] Chen Gong and Yvon Maday. Directed Message Passing Based on Attention for Prediction of Molecular Properties. *arXiv preprint arXiv:2305.14819*, 2023. URL <https://arxiv.org/abs/2305.14819>.
- [76] Rogerio Almeida Gouvea, Pierre-Paul De Breuck, Tatiane Pretto, Gian-Marco Rignanes, and Marcos Jose Leite Santos. Combining feature-based approaches with graph neural networks and symbolic regression for synergistic performance and interpretability. *arXiv preprint arXiv:2509.03547*, 2025. URL <https://arxiv.org/abs/2509.03547>.
- [77] Vitor F. Grizzi, Luke N. Pretzie, Jiayi Xu, and Cong Liu. XANE(3): An E(3)-Equivariant Graph Neural Network for Accurate Prediction of XANES Spectra from Atomic Structures. *arXiv preprint arXiv:2604.12140*, 2026. URL <https://arxiv.org/abs/2604.12140>.
- [78] Cameron Gruich, Varun Madhavan, Yixin Wang, and Bryan Goldsmith. Clarifying Trust of Materials Property Predictions using Neural Networks with Distribution-Specific Uncertainty Quantification. *arXiv preprint arXiv:2302.02595*, 2023. URL <https://arxiv.org/abs/2302.02595>.
- [79] Davy Guan, Lu Zhang, Asiri Wijesinghe, Allen Zhu, He Zhao, Helen Power, F. Hafna Ahmed, Andrew Warden, Cheng Soon Ong, and Daniel M. Steinberg. Can Tabular In-Context Learners Generalize to Biomolecular Property Prediction? *arXiv preprint arXiv:2606.31126*, 2026. URL <https://arxiv.org/abs/2606.31126>.
- [80] Jinjiang Guo and Sheng Ding. Do Larger Models Really Win in Drug Discovery? A Benchmark Assessment of Model Scaling in AI-Driven Molecular Property and Activity Prediction. *arXiv preprint arXiv:2604.26498*, 2026. URL <https://arxiv.org/abs/2604.26498>.
- [81] Abhijit Gupta. CardiGraphormer: Unveiling the Power of Self-Supervised Learning in Revolutionizing Drug Discovery. *arXiv preprint arXiv:2307.00859*, 2023. URL <https://arxiv.org/abs/2307.00859>.
- [82] Jiaqi Han, Yu Rong, Tingyang Xu, and Wenbing Huang. Geometrically Equivariant Graph Neural Networks: A Survey. *arXiv preprint arXiv:2202.07230*, 2022. URL <https://arxiv.org/abs/2202.07230>.
- [83] Qiwei Han, Chi Zhou, Ruobing Wang, and Zheng Ma. Multimodal Molecular Representation Learning with Graph Neural Networks, Deep & Cross Networks, and SMILES Embeddings. *arXiv preprint arXiv:2607.05736*, 2026. URL <https://arxiv.org/abs/2607.05736>.
- [84] Xiao-Qi Han, Ze-Feng Gao, Wen-Kao Li, Peng-Jie Guo, and Zhong-Yi Lu. PhononBench: A Large-Scale Phonon-Based Benchmark for Dynamical

- Stability in Crystal Generation. *arXiv preprint arXiv:2512.21227*, 2025. URL <https://arxiv.org/abs/2512.21227>.
- [85] Hasi Hays and William J. Richardson. RAG-GNN: Integrating Retrieved Knowledge with Graph Neural Networks for Precision Medicine. *arXiv preprint arXiv:2602.00586*, 2026. URL <https://arxiv.org/abs/2602.00586>.
- [86] Yinhan He, Zaiyi Zheng, Patrick Soga, Yaozhen Zhu, yushun Dong, and Jundong Li. Explaining Graph Neural Networks with Large Language Models: A Counterfactual Perspective for Molecular Property Prediction. *arXiv preprint arXiv:2410.15165*, 2024. URL <https://arxiv.org/abs/2410.15165>.
- [87] Hatem Helal, Jesun Firoz, Jenna Bilbrey, Mario Michael Krell, Tom Murray, Ang Li, Sotiris Xantheas, and Sutanay Choudhury. Extreme Acceleration of Graph Neural Network-based Prediction Models for Quantum Chemistry. *arXiv preprint arXiv:2211.13853*, 2022. URL <https://arxiv.org/abs/2211.13853>.
- [88] Franziska Herbert, Vignesh Prasad, Han Liu, Dorothea Koert, and Georgia Chalvatzaki. Semantic-Geometric Task Representations for Bimanual Manipulation from Human Demonstrations to Robot Action Planning. *arXiv preprint arXiv:2601.11460*, 2026. URL <https://arxiv.org/abs/2601.11460>.
- [89] Huiyang Hong, Xinkai Wu, Hongyu Sun, Chaoyang Xie, Qi Wang, and Yuquan Li. Learning Hierarchical Interaction for Accurate Molecular Property Prediction. *arXiv preprint arXiv:2504.20127*, 2025. URL <https://arxiv.org/abs/2504.20127>.
- [90] Ryien Hosseini, Filippo Simini, Austin Clyde, and Arvind Ramanathan. Deep Surrogate Docking: Accelerating Automated Drug Discovery with Graph Neural Networks. *arXiv preprint arXiv:2211.02720*, 2022. URL <https://arxiv.org/abs/2211.02720>.
- [91] Xiaoyang Hou, Tian Zhu, Milong Ren, Dongbo Bu, Xin Gao, Chunming Zhang, and Shiwei Sun. Improving Molecular Graph Generation with Flow Matching and Optimal Transport. *arXiv preprint arXiv:2411.05676*, 2024. URL <https://arxiv.org/abs/2411.05676>.
- [92] Circe Hsu, Claire Schlesinger, Karan Mudaliar, Jordan Leung, Robin Walters, and Peter Schindler. FIRE-GNN: Force-informed, Relaxed Equivariance Graph Neural Network for Rapid and Accurate Prediction of Surface Properties. *arXiv preprint arXiv:2508.16012*, 2025. URL <https://arxiv.org/abs/2508.16012>.
- [93] HaoWei Hua, Chen Liang, Ding Pan, Irwin King, Shengchao Liu, Koji Tsuda, and Wanyu Lin. Scalable Dielectric Tensor Predictions for Inorganic Materials using Equivariant Graph Neural Networks. *arXiv preprint arXiv:2601.04755*, 2026. URL <https://arxiv.org/abs/2601.04755>.
- [94] Han Huang, Leilei Sun, Bowen Du, and Weifeng Lv. Learning Joint 2D & 3D Diffusion Models for Complete Molecule Generation. *arXiv preprint arXiv:2305.12347*, 2023. URL <https://arxiv.org/abs/2305.12347>.
- [95] Ningyuan Huang, Richard Stiskalek, Jun-Young Lee, Adrian E. Bayer, Charles C. Margosian, Christian Kragh Jespersen, Lucia A. Perez, Lawrence K. Saul, and Francisco Villaseca-Navarro. CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning. *arXiv preprint arXiv:2507.03707*, 2025. URL <https://arxiv.org/abs/2507.03707>.
- [96] Xiaobao Huang, Yihong Ma, Anjali Gurajapu, Jules Schleinitz, Zhichun Guo, Sarah E. Reisman, and Nitesh V. Chawla. ChemHGNN: A Hierarchical Hypergraph Neural Network for Reaction Virtual Screening and Discovery. *arXiv preprint arXiv:2506.11041*, 2025. URL <https://arxiv.org/abs/2506.11041>.
- [97] Yinan Huang, Xingang Peng, Jianzhu Ma, and Muhao Zhang. Boosting the Cycle Counting Power of Graph Neural Networks with I²S-GNNs. *arXiv preprint arXiv:2210.13978*, 2022. URL <https://arxiv.org/abs/2210.13978>.
- [98] Achintha Ihalage and Yang Hao. Formula graph self-attention network for representation-domain independent materials discovery. *arXiv preprint arXiv:2201.05649*, 2022. URL <https://arxiv.org/abs/2201.05649>.
- [99] Katsuhiko Ishiguro, Shin ichi Maeda, and Masanori Koyama. Graph Warp Module: an Auxiliary Module for Boosting the Power of Graph Neural Networks in Molecular Graph Analysis. *arXiv preprint arXiv:1902.01020*, 2019. URL <https://arxiv.org/abs/1902.01020>.
- [100] Shaoxiong Ji, Ya Gao, and Pekka Marttinen. Knowledge-augmented Graph Neural Networks with Concept-aware Attention for Adverse Drug Event Detection. *arXiv preprint arXiv:2301.10451*, 2023. URL <https://arxiv.org/abs/2301.10451>.
- [101] Yuanfeng Ji, Lu Zhang, Jiaxiang Wu, Bingzhe Wu, Long-Kai Huang, Tingyang Xu, Yu Rong, Lanqing Li, Jie Ren, Ding Xue, Houtim Lai, Shaoyong Xu, Jing Feng, Wei Liu, Ping Luo, Shuigeng Zhou, Junzhou Huang, Peilin Zhao, and Yatao Bian. DrugOOD: Out-of-Distribution (OOD) Dataset Curator and Benchmark for AI-aided Drug Discovery – A Focus on Affinity Prediction Problems with Noise Annotations. *arXiv preprint*

- arXiv:2201.09637*, 2022. URL <https://arxiv.org/abs/2201.09637>.
- [102] Yue Jian, Yuyang Wang, and Amir Barati Farmani. Predicting CO₂ Absorption in Ionic Liquids with Molecular Descriptors and Explainable Graph Neural Networks. *arXiv preprint arXiv:2210.01120*, 2022. URL <https://arxiv.org/abs/2210.01120>.
- [103] Shengli Jiang and Prasanna Balaprakash. Graph Neural Network Architecture Search for Molecular Property Prediction. *arXiv preprint arXiv:2008.12187*, 2020. URL <https://arxiv.org/abs/2008.12187>.
- [104] Tianyi Jiang, Zeyu Wang, Shanqing Yu, and Qi Xuan. Adaptive Substructure-Aware Expert Model for Molecular Property Prediction. *arXiv preprint arXiv:2504.05844*, 2025. URL <https://arxiv.org/abs/2504.05844>.
- [105] Dian Jin, Yancheng Yuan, and Xiaoming Tao. Magnitude-Modulated Equivariant Adapter for Parameter-Efficient Fine-Tuning of Equivariant Graph Neural Networks. *arXiv preprint arXiv:2511.06696*, 2025. URL <https://arxiv.org/abs/2511.06696>.
- [106] Jeonghee Jo, Bumju Kwak, Byunghan Lee, and Sungroh Yoon. Flexible dual-branched message passing neural network for quantum mechanical property prediction with molecular conformation. *arXiv preprint arXiv:2106.07273*, 2021. URL <https://arxiv.org/abs/2106.07273>.
- [107] Peter Bjørn Jrgensen and Arghya Bhowmik. Equivariant graph neural networks for fast electron density estimation of molecules, liquids, and solids. *arXiv preprint arXiv:2112.00652*, 2021. URL <https://arxiv.org/abs/2112.00652>.
- [108] Sekou-Oumar Kaba and Siamak Ravanbakhsh. Equivariant Networks for Crystal Structures. *arXiv preprint arXiv:2211.15420*, 2022. URL <https://arxiv.org/abs/2211.15420>.
- [109] Jaykumar Kakkad, Jaspal Jannu, Kartik Sharma, Charu Aggarwal, and Sourav Medya. A Survey on Explainability of Graph Neural Networks. *arXiv preprint arXiv:2306.01958*, 2023. URL <https://arxiv.org/abs/2306.01958>.
- [110] Sidharth Kannan, Pooyan Goodarzi, Evangelos E. Papalexakis, and Jonathan W. Richardson. Graph Neural Networks for Interferometer Simulations. *arXiv preprint arXiv:2512.16051*, 2025. URL <https://arxiv.org/abs/2512.16051>.
- [111] Prathik R Kaundinya, Kamal Choudhary, and Surya R. Kalidindi. Prediction of the electron density of states for crystalline compounds with Atomistic Line Graph Neural Networks (ALIGNN). *arXiv preprint arXiv:2201.08348*, 2022. URL <https://arxiv.org/abs/2201.08348>.
- [112] Alexander Kensert, Gert Desmet, and Deirdre Cabooter. MolGraph: a Python package for the implementation of molecular graphs and graph neural networks with TensorFlow and Keras. *arXiv preprint arXiv:2208.09944*, 2022. URL <https://arxiv.org/abs/2208.09944>.
- [113] Rahul Khorana. Topological Feature Compression for Molecular Graph Neural Networks. *arXiv preprint arXiv:2508.07807*, 2025. URL <https://arxiv.org/abs/2508.07807>.
- [114] Hyeonsu Kim, Jeheon Woo, Seonghwan Kim, Seokhyun Moon, Jun Hyeong Kim, and Woo Youn Kim. GeoTMI: Predicting quantum chemical property with easy-to-obtain geometry via positional denoising. *arXiv preprint arXiv:2304.03724*, 2023. URL <https://arxiv.org/abs/2304.03724>.
- [115] Akihiro Kishimoto, Hiroshi Kajino, Masataka Hirose, Junta Fuchiwaki, Indra Priyadarsini, Lisa Hamada, Hajime Shinohara, Daiju Nakano, and Seiji Takeda. MHG-GNN: Combination of Molecular Hypergraph Grammar with Graph Neural Network. *arXiv preprint arXiv:2309.16374*, 2023. URL <https://arxiv.org/abs/2309.16374>.
- [116] Madina Kojanazarova, Sidaty El Hadramy, Jack Wilkie, Georg Rauter, and Philippe C. Cattin. Soft Tissue Simulation and Force Estimation from Heterogeneous Structures using Equivariant Graph Neural Networks. *arXiv preprint arXiv:2509.10125*, 2025. URL <https://arxiv.org/abs/2509.10125>.
- [117] Jian-Gang Kong, Qing-Xu Li, Jian Li, Yu Liu, and Jia-Ji Zhu. Self-supervised graph neural networks for accurate prediction of Neel temperature. *arXiv preprint arXiv:2206.04109*, 2022. URL <https://arxiv.org/abs/2206.04109>.
- [118] Vadim Korolev and Artem Mitrofanov. Coarse-grained crystal graph neural networks for reticular materials design. *arXiv preprint arXiv:2310.19500*, 2023. URL <https://arxiv.org/abs/2310.19500>.
- [119] Edward Elson Kosasih, Joaquin Cabezas, Xavier Sumba, Piotr Bielak, Kamil Tagowski, Kelvin Idanwekhai, Benedict Aaron Tjandra, and Arrian Rokkum Jamasb. On Graph Neural Network Ensembles for Large-Scale Molecular Property Prediction. *arXiv preprint arXiv:2106.15529*, 2021. URL <https://arxiv.org/abs/2106.15529>.
- [120] Jonathan Krook, Axel Janson, Joakim Anden, Melanie Weber, and Ozan Oktem. Protein Graph Neural Networks for Heterogeneous Cryo-EM Reconstruction. *arXiv preprint arXiv:2602.21915*, 2026. URL <https://arxiv.org/abs/2602.21915>.
- [121] George Lamb and Brooks Paige. Bayesian Graph Neural Networks for Molecular Property Prediction.

- tion. *arXiv preprint arXiv:2012.02089*, 2020. URL <https://arxiv.org/abs/2012.02089>.
- [122] Yi-Shan Lan, Pin-Yu Chen, and Tsung-Yi Ho. NaNa and MiGu: Semantic Data Augmentation Techniques to Enhance Protein Classification in Graph Neural Networks. *arXiv preprint arXiv:2403.14736*, 2024. URL <https://arxiv.org/abs/2403.14736>.
- [123] Leo Laugier, Daniil Bash, Jose Recatala, Hong Kuan Ng, Savitha Ramasamy, Chuan-Sheng Foo, Vijay R Chandrasekhar, and Kedar Hippalgaonkar. Predicting thermoelectric properties from crystal graphs and material descriptors - first application for functional materials. *arXiv preprint arXiv:1811.06219*, 2018. URL <https://arxiv.org/abs/1811.06219>.
- [124] Tuan Le, Frank Noe, and Djork-Arne Clevert. Equivariant Graph Attention Networks for Molecular Property Prediction. *arXiv preprint arXiv:2202.09891*, 2022. URL <https://arxiv.org/abs/2202.09891>.
- [125] Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary. AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org. *arXiv preprint arXiv:2512.11935*, 2025. URL <https://arxiv.org/abs/2512.11935>.
- [126] Chaojian Li, Zhifan Ye, Massimiliano Lupo Pasini, Jong Youl Choi, Cheng Wan, Yingyan Celine Lin, and Prasanna Balaprakash. Scaling Laws of Graph Neural Networks for Atomistic Materials Modeling. *arXiv preprint arXiv:2504.08112*, 2025. URL <https://arxiv.org/abs/2504.08112>.
- [127] Han Li, Dan Zhao, and Jianyang Zeng. KPGT: Knowledge-Guided Pre-training of Graph Transformer for Molecular Property Prediction. *arXiv preprint arXiv:2206.03364*, 2022. URL <https://arxiv.org/abs/2206.03364>.
- [128] Haoyang Li, Yuming Xu, Alexander Zhou, Yongqi Zhang, Jason Chen Zhang, Lei Chen, and Qing Li. OpenGLT: A Comprehensive Benchmark of Graph Neural Networks for Graph-Level Tasks. *arXiv preprint arXiv:2501.00773*, 2025. URL <https://arxiv.org/abs/2501.00773>.
- [129] Kun Li, Zhennan Wu, Yida Xiong, Hongzhi Zhang, Longtao Hu, Zhonglie Liu, Junqi Zeng, Wenjie Wu, Mukun Chen, Jiameng Chen, and Wenbin Hu. BSL: A Unified and Generalizable Multitask Learning Platform for Virtual Drug Discovery from Design to Synthesis. *arXiv preprint arXiv:2508.01195*, 2025. URL <https://arxiv.org/abs/2508.01195>.
- [130] Longlong Li, Yipeng Zhang, Guanghui Wang, and Kelin Xia. KA-GNN: Kolmogorov-Arnold Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2410.11323*, 2024. URL <https://arxiv.org/abs/2410.11323>.
- [131] Mufei Li, Jinjing Zhou, Jiajing Hu, Wenxuan Fan, Yangkang Zhang, Yaxin Gu, and George Karypis. DGL-LifeSci: An Open-Source Toolkit for Deep Learning on Graphs in Life Science. *arXiv preprint arXiv:2106.14232*, 2021. URL <https://arxiv.org/abs/2106.14232>.
- [132] Shuangli Li, Jingbo Zhou, Tong Xu, Dejing Dou, and Hui Xiong. GeomGCL: Geometric Graph Contrastive Learning for Molecular Property Prediction. *arXiv preprint arXiv:2109.11730*, 2021. URL <https://arxiv.org/abs/2109.11730>.
- [133] Zian Li, Xiyuan Wang, Yinan Huang, and Muhan Zhang. Is Distance Matrix Enough for Geometric Deep Learning? *arXiv preprint arXiv:2302.05743*, 2023. URL <https://arxiv.org/abs/2302.05743>.
- [134] Chen Liang, Bowen Wang, Shaogang Hao, Guangyong Chen, Pheng-Ann Heng, and Xiaolong Zou. Multi-Task Mixture Density Graph Neural Networks for Predicting Cu-based Single-Atom Alloy Catalysts for CO₂ Reduction Reaction. *arXiv preprint arXiv:2209.07300*, 2022. URL <https://arxiv.org/abs/2209.07300>.
- [135] Youwei Liang, Ruiyi Zhang, Li Zhang, and Pengtao Xie. DrugChat: Towards Enabling ChatGPT-Like Capabilities on Drug Molecule Graphs. *arXiv preprint arXiv:2309.03907*, 2023. URL <https://arxiv.org/abs/2309.03907>.
- [136] Peiyuan Liao, Han Zhao, Keyulu Xu, Tommi Jaakkola, Geoffrey Gordon, Stefanie Jegelka, and Ruslan Salakhutdinov. Information Obfuscation of Graph Neural Networks. *arXiv preprint arXiv:2009.13504*, 2020. URL <https://arxiv.org/abs/2009.13504>.
- [137] Chang Liu, Vivian Li, Linus Leong, Vladimir Radenkovic, Pietro Lio, and Chaitanya K. Joshi. Multi-Scale Protein Structure Modelling with Geometric Graph U-Nets. *arXiv preprint arXiv:2512.06752*, 2025. URL <https://arxiv.org/abs/2512.06752>.
- [138] Gang Liu, Michael Sun, Wojciech Matusik, Meng Jiang, and Jie Chen. Multimodal Large Language Models for Inverse Molecular Design with Retrosynthetic Planning. *arXiv preprint arXiv:2410.04223*, 2024. URL <https://arxiv.org/abs/2410.04223>.
- [139] Meng Liu, Cong Fu, Xuan Zhang, Limei Wang, Yaochen Xie, Hao Yuan, Youzhi Luo, Zhao Xu, Shenglong Xu, and Shuiwang Ji. Fast Quantum Property Prediction via Deeper 2D and 3D Graph Networks. *arXiv preprint arXiv:2106.08551*, 2021. URL <https://arxiv.org/abs/2106.08551>.
- [140] XiaYu Liu, Chao Fan, Yang Liu, and Hou biao

- Li. Multi-Level Fusion Graph Neural Network for Molecule Property Prediction. *arXiv preprint arXiv:2507.03430*, 2025. URL <https://arxiv.org/abs/2507.03430>.
- [141] Yujie Liu, Zhenyu Wang, Hang Lei, Guoyu Zhang, Jiawei Xian, Zhibin Gao, Jun Sun, Haifeng Song, and Xiangdong Ding. Machine learning-driven elasticity prediction in advanced inorganic materials via convolutional neural networks. *arXiv preprint arXiv:2511.04468*, 2025. URL <https://arxiv.org/abs/2511.04468>.
- [142] Valentin Lombard, Julien Nguyen Van, Sergei Grudin, and Elodie Laine. PETIMOT: A Novel Framework for Inferring Protein Motions from Sparse Data Using SE(3)-Equivariant Graph Neural Networks. *arXiv preprint arXiv:2504.02839*, 2025. URL <https://arxiv.org/abs/2504.02839>.
- [143] Kha-Dinh Luong and Ambuj Singh. Fragment-based Pretraining and Finetuning on Molecular Graphs. *arXiv preprint arXiv:2310.03274*, 2023. URL <https://arxiv.org/abs/2310.03274>.
- [144] Hehuan Ma, Yatao Bian, Yu Rong, Wenbing Huang, Tingyang Xu, Weiyang Xie, Geyan Ye, and Junzhou Huang. Multi-View Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2005.13607*, 2020. URL <https://arxiv.org/abs/2005.13607>.
- [145] Ali Madani, Cyna Shirazinejad, Jia Rui Ong, Hengameh Shams, and Mohammad Mofrad. Pro-Dyn0: Inferring calponin homology domain stretching behavior using graph neural networks. *arXiv preprint arXiv:1910.09738*, 2019. URL <https://arxiv.org/abs/1910.09738>.
- [146] Sourav Mal and Satadeep Bhattacharjee. mCGCNN: A Dual-Stream Crystal Graph Convolutional Neural Network for the Efficient Prediction of Magnetic Properties of Crystalline Materials. *arXiv preprint arXiv:2606.28458*, 2026. URL <https://arxiv.org/abs/2606.28458>.
- [147] Adil Mudasir Malla and Asif Ali Banka. A Systematic Review of Deep Graph Neural Networks: Challenges, Classification, Architectures, Applications & Potential Utility in Bioinformatics. *arXiv preprint arXiv:2311.02127*, 2023. URL <https://arxiv.org/abs/2311.02127>.
- [148] Dominic Masters, Josef Dean, Kerstin Klaser, Zhiyi Li, Sam Maddrell-Mander, Adam Sanders, Hatem Helal, Deniz Beker, Andrew Fitzgibbon, Shenyang Huang, Ladislav Rampasek, and Dominique Beaini. GPS++: Reviving the Art of Message Passing for Molecular Property Prediction. *arXiv preprint arXiv:2302.02947*, 2023. URL <https://arxiv.org/abs/2302.02947>.
- [149] Nicholas Miklaucic, Lai Wei, Rongzhi Dong, Nihang Fu, Sadman Sadeed Ome, Qingyang Li, Sourin Dey, Victor Fung, and Jianjun Hu. Facet: highly efficient E(3)-equivariant networks for interatomic potentials. *arXiv preprint arXiv:2509.08418*, 2025. URL <https://arxiv.org/abs/2509.08418>.
- [150] Joshua Mitton, Hans M. Senn, Klaas Wynne, and Roderick Murray-Smith. A Graph VAE and Graph Transformer Approach to Generating Molecular Graphs. *arXiv preprint arXiv:2104.04345*, 2021. URL <https://arxiv.org/abs/2104.04345>.
- [151] Alex Morehead and Jianlin Cheng. Geometry-Complete Diffusion for 3D Molecule Generation and Optimization. *arXiv preprint arXiv:2302.04313*, 2023. URL <https://arxiv.org/abs/2302.04313>.
- [152] Shrimon Mukherjee, Madhusudan Ghosh, and Partha Basuchowdhuri. CrysAtom: Distributed Representation of Atoms for Crystal Property Prediction. *arXiv preprint arXiv:2409.04737*, 2024. URL <https://arxiv.org/abs/2409.04737>.
- [153] Shrimon Mukherjee, Kishalay Das, Partha Basuchowdhuri, Pawan Goyal, and Niloy Ganguly. Latent Diffusion Pretraining for Crystal Property Prediction. *arXiv preprint arXiv:2606.00776*, 2026. URL <https://arxiv.org/abs/2606.00776>.
- [154] Sebastian Musia, Bartosz Zielinski, and Tomasz Danel. Fragment-Wise Interpretability in Graph Neural Networks via Molecule Decomposition and Contribution Analysis. *arXiv preprint arXiv:2508.15015*, 2025. URL <https://arxiv.org/abs/2508.15015>.
- [155] Joseph Musielewicz, Janice Lan, Matt Uyttendaele, and John R. Kitchin. Improved Uncertainty Estimation of Graph Neural Network Potentials Using Engineered Latent Space Distances. *arXiv preprint arXiv:2407.10844*, 2024. URL <https://arxiv.org/abs/2407.10844>.
- [156] Ehsan Naghavi, Haifeng Wang, Vahid Ziaei Rad, Julius Guccione, Ghassan Kassab, Vishnu Boddeti, Seungik Baek, and Lik-Chuan Lee. Rapid prediction of cardiac activation in the left ventricle with geometric deep learning: a step towards cardiac resynchronization therapy planning. *arXiv preprint arXiv:2506.08987*, 2025. URL <https://arxiv.org/abs/2506.08987>.
- [157] Marlen Neubert, Patrick Reiser, Frauke Grater, and Pascal Friederich. Learning Potential Energy Surfaces of Hydrogen Atom Transfer Reactions in Peptides. *arXiv preprint arXiv:2508.00578*, 2025. URL <https://arxiv.org/abs/2508.00578>.
- [158] Cuong Q. Nguyen, Constantine Kretsoulas, and Kim M. Branson. Meta-Learning GNN Initializations for Low-Resource Molecular Property Prediction. *arXiv preprint arXiv:2003.05996*, 2020. URL <https://arxiv.org/abs/2003.05996>.

- [159] Emily Nguyen, Yongchan Hong, Harsh Toshniwal, Yan Liu, and Andreas Lutten. GLACIER: A Multimodal Student-Teacher Foundation Model for Molecular Property Prediction. *arXiv preprint arXiv:2606.11382*, 2026. URL <https://arxiv.org/abs/2606.11382>.
- [160] Ngoc-Quang Nguyen. EquiCPI: SE(3)-Equivariant Geometric Deep Learning for Structure-Aware Prediction of Compound-Protein Interactions. *arXiv preprint arXiv:2504.04654*, 2025. URL <https://arxiv.org/abs/2504.04654>.
- [161] Thao Nguyen, Kuan-Hao Huang, Ge Liu, Martin D. Burke, Ying Diao, and Heng Ji. FARM: Enhancing Molecular Representations with Functional Group Awareness. *arXiv preprint arXiv:2410.02082*, 2024. URL <https://arxiv.org/abs/2410.02082>.
- [162] Thao Nguyen, Tiara Torres-Flores, Changhyun Hwang, Carl Edwards, Ying Diao, and Heng Ji. GLaD: Synergizing Molecular Graphs and Language Descriptors for Enhanced Power Conversion Efficiency Prediction in Organic Photovoltaic Devices. *arXiv preprint arXiv:2405.14203*, 2024. URL <https://arxiv.org/abs/2405.14203>.
- [163] Trung Nguyen and Duc Duy Nguyen. MMGNN: Multi-level, multi-color graph neural networks for molecular property prediction. *arXiv preprint arXiv:2606.20906*, 2026. URL <https://arxiv.org/abs/2606.20906>.
- [164] Trung Nguyen, Md Masud Rana, Farjana Tasnim Mukta, Chang-Guo Zhan, and Duc Duy Nguyen. Geometric Multi-color Message Passing Graph Neural Networks for Blood-brain Barrier Permeability Prediction. *arXiv preprint arXiv:2507.18926*, 2025. URL <https://arxiv.org/abs/2507.18926>.
- [165] Jianan Nie and Peng Gao. FlowMS: Flow Matching for De Novo Structure Elucidation from Mass Spectra. *arXiv preprint arXiv:2603.18397*, 2026. URL <https://arxiv.org/abs/2603.18397>.
- [166] Idelfonso B. R. Nogueira, Carine M. Rebello, Mumin Enis Leblebici, and Erick Giovanni Sperandio Nascimento. MultiPUFFIN: A Multimodal Domain-Constrained Foundation Model for Molecular Property Prediction of Small Molecules. *arXiv preprint arXiv:2603.00857*, 2026. URL <https://arxiv.org/abs/2603.00857>.
- [167] Janghoon Ock, Srivathsan Badrinarayanan, Rishikesh Magar, Akshay Antony, and Amir Barati Farimani. Multimodal Language and Graph Learning of Adsorption Configuration in Catalysis. *arXiv preprint arXiv:2401.07408*, 2024. URL <https://arxiv.org/abs/2401.07408>.
- [168] Sadman Sadeed Omeed, Steph-Yves Louis, Nihang Fu, Lai Wei, Sourin Dey, Rongzhi Dong, Qinyang Li, and Jianjun Hu. Scalable deeper graph neural networks for high-performance materials property prediction. *arXiv preprint arXiv:2109.12283*, 2021. URL <https://arxiv.org/abs/2109.12283>.
- [169] Sadman Sadeed Omeed, Nihang Fu, Rongzhi Dong, Ming Hu, and Jianjun Hu. Structure-based out-of-distribution (OOD) materials property prediction: a benchmark study. *arXiv preprint arXiv:2401.08032*, 2024. URL <https://arxiv.org/abs/2401.08032>.
- [170] Hakan T. Ota, Abdulhamit Subasi, Furkan Kurt, M. Abdullah Canbaz, and Yasin Uzun. Analysis of Gene Regulatory Networks from Gene Expression Using Graph Neural Networks. *arXiv preprint arXiv:2409.13664*, 2024. URL <https://arxiv.org/abs/2409.13664>.
- [171] Teerachote Pakornchote, Annap Ektarawong, and Thiparat Chotibut. StrainTensorNet: Predicting crystal structure elastic properties using SE(3)-equivariant graph neural networks. *arXiv preprint arXiv:2306.12818*, 2023. URL <https://arxiv.org/abs/2306.12818>.
- [172] Cheol Woo Park and Chris Wolverton. Developing an improved Crystal Graph Convolutional Neural Network framework for accelerated materials discovery. *arXiv preprint arXiv:1906.05267*, 2019. URL <https://arxiv.org/abs/1906.05267>.
- [173] Yutack Park, Jaesun Kim, Seungwoo Hwang, and Seungwu Han. Scalable Parallel Algorithm for Graph Neural Network Interatomic Potentials in Molecular Dynamics Simulations. *arXiv preprint arXiv:2402.03789*, 2024. URL <https://arxiv.org/abs/2402.03789>.
- [174] Rushikesh Pawar, Harshit Rawat, Ayush Kumar, and Phani Motamarri. Prototype-Guided Latent Alignment for Data-Efficient Fine-Tuning of Molecular Foundation Models. *arXiv preprint arXiv:2605.29969*, 2026. URL <https://arxiv.org/abs/2605.29969>.
- [175] Jiayu Peng, James Damewood, Jessica Karaguegian, Jaclyn R. Lurger, and Rafael Gomez-Bombarelli. Learning Ordering in Crystalline Materials with Symmetry-Aware Graph Neural Networks. *arXiv preprint arXiv:2409.13851*, 2024. URL <https://arxiv.org/abs/2409.13851>.
- [176] Zihan Pengmei, Chatipat Lorpai boon, Spencer C. Guo, Jonathan Weare, and Aaron R. Dinner. Using pretrained graph neural networks with token mixers as geometric featurizers for conformational dynamics. *arXiv preprint arXiv:2409.19838*, 2024. URL <https://arxiv.org/abs/2409.19838>.
- [177] Oleg Platonov, Gleb Bazhenov, Dmitry Ereemeev, and Liudmila Prokhorenkova. A Fair Evaluation of Graph Foundation Models for Node Property

- Prediction. *arXiv preprint arXiv:2606.24509*, 2026. URL <https://arxiv.org/abs/2606.24509>.
- [178] Can Polat, Hasan Kurban, Erchin Serpedin, and Mustafa Kurban. Understanding the Capabilities of Molecular Graph Neural Networks in Materials Science Through Multimodal Learning and Physical Context Encoding. *arXiv preprint arXiv:2505.12137*, 2025. URL <https://arxiv.org/abs/2505.12137>.
- [179] Can Polat, Mehmet Tuncel, Mustafa Kurban, Erchin Serpedin, and Hasan Kurban. xChemAgents: Agentic AI for Explainable Quantum Chemistry. *arXiv preprint arXiv:2505.20574*, 2025. URL <https://arxiv.org/abs/2505.20574>.
- [180] Peerasait Prachaseree and Emma Lejeune. Learning Mechanically Driven Emergent Behavior with Message Passing Neural Networks. *arXiv preprint arXiv:2202.01380*, 2022. URL <https://arxiv.org/abs/2202.01380>.
- [181] M V Sai Prakash, Siddhartha Reddy N, Ganesh Parab, Varun V, Vishal Vaddina, and Saisubramaniam Gopalakrishnan. Synergistic Fusion of Graph and Transformer Features for Enhanced Molecular Property Prediction. *arXiv preprint arXiv:2310.03027*, 2023. URL <https://arxiv.org/abs/2310.03027>.
- [182] Magdalena Proszewska, Tomasz Danel, and Dawid Rymarczyk. B-XAIC Dataset: Benchmarking Explainable AI for Graph Neural Networks Using Chemical Data. *arXiv preprint arXiv:2505.22252*, 2025. URL <https://arxiv.org/abs/2505.22252>.
- [183] Chen Qian, Yunhai Xiong, and Xiang Chen. Directed Graph Attention Neural Network Utilizing 3D Coordinates for Molecular Property Prediction. *arXiv preprint arXiv:2012.00404*, 2020. URL <https://arxiv.org/abs/2012.00404>.
- [184] Zhuoran Qiao, Anders S. Christensen, Matthew Welborn, Frederick R. Manby, Anima Anandkumar, and Thomas F. Miller. Informing Geometric Deep Learning with Electronic Interactions to Accelerate Quantum Chemistry. *arXiv preprint arXiv:2105.14655*, 2021. URL <https://arxiv.org/abs/2105.14655>.
- [185] Jiangjie Qiu, Hou Hei Lam, Xiuyuan Hu, Wentao Li, Siwei Fu, Fankun Zeng, Hao Zhang, and Xiaonan Wang. Accelerating High-Efficiency Organic Photovoltaic Discovery via Pretrained Graph Neural Networks and Generative Reinforcement Learning. *arXiv preprint arXiv:2503.23766*, 2025. URL <https://arxiv.org/abs/2503.23766>.
- [186] Chowdhury Mohammad Abid Rahman, Ghadendra Bhandari, Nasser M Nasrabadi, Aldo H. Romero, and Prashna K. Gyawali. Enhancing material property prediction with ensemble deep graph convolutional networks. *arXiv preprint arXiv:2407.18847*, 2024. URL <https://arxiv.org/abs/2407.18847>.
- [187] Md Habibur Rahman, Prince Gollapalli, Panayotis Manganaris, Satyesh Kumar Yadav, Ghanshyam Pilania, Brian DeCost, Kamal Choudhary, and Arun Mannodi-Kanakkithodi. Accelerating Defect Predictions in Semiconductors Using Graph Neural Networks. *arXiv preprint arXiv:2309.06423*, 2023. URL <https://arxiv.org/abs/2309.06423>.
- [188] Snehal Raj, Brian Coyle, Leo Monbroussou, Andre J. Ferreira-Martins, Renato M. S. Farias, and Elham Kashefi. Scalable Message-Passing Quantum Graph Neural Networks in the Weisfeiler-Leman Hierarchy. *arXiv preprint arXiv:2606.26873*, 2026. URL <https://arxiv.org/abs/2606.26873>.
- [189] Jiahua Rao, Shuangjia Zheng, and Yuedong Yang. Quantitative Evaluation of Explainable Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2107.04119*, 2021. URL <https://arxiv.org/abs/2107.04119>.
- [190] Neel Redkar. CLAMP: A Contrastive Language And Molecule Pre-training Network. *arXiv preprint arXiv:2311.07617*, 2023. URL <https://arxiv.org/abs/2311.07617>.
- [191] Janosh Riebesell, Rhys E. A. Goodall, Philipp Benner, Yuan Chiang, Bowen Deng, Gerbrand Ceder, Mark Asta, Alpha A. Lee, Anubhav Jain, and Kristin A. Persson. Matbench Discovery – A framework to evaluate machine learning crystal stability predictions. *arXiv preprint arXiv:2308.14920*, 2023. URL <https://arxiv.org/abs/2308.14920>.
- [192] Oscar Rivera, Ziqing Wang, Matthieu Dagommer, Abhishek Pandey, and Kaize Ding. GlassMol: Interpretable Molecular Property Prediction with Concept Bottleneck Models. *arXiv preprint arXiv:2603.01274*, 2026. URL <https://arxiv.org/abs/2603.01274>.
- [193] Yu Rong, Yatao Bian, Tingyang Xu, Weiyang Xie, Ying Wei, Wenbing Huang, and Junzhou Huang. Self-Supervised Graph Transformer on Large-Scale Molecular Data. *arXiv preprint arXiv:2007.02835*, 2020. URL <https://arxiv.org/abs/2007.02835>.
- [194] Ivan Rubtsov, Ivan Dudakov, Yuri Kuratov, and Vadim Korolev. Enhancing composition-based materials property prediction by cross-modal knowledge transfer. *arXiv preprint arXiv:2511.03371*, 2025. URL <https://arxiv.org/abs/2511.03371>.
- [195] Robin Ruff, Patrick Reiser, Jan Stuhmer, and Pascal Friederich. Connectivity Optimized Nested Graph Networks for Crystal Structures. *arXiv preprint arXiv:2302.14102*, 2023. URL <https://arxiv.org/abs/2302.14102>.
- [196] Serim Ryou, Michael R. Maser, Alexander Y. Cui,

- Travis J. DeLano, Yisong Yue, and Sarah E. Reisman. Graph Neural Networks for the Prediction of Substrate-Specific Organic Reaction Conditions. *arXiv preprint arXiv:2007.04275*, 2020. URL <https://arxiv.org/abs/2007.04275>.
- [197] Nader Sadek, Mirette Moawad, Christina Naguib, and Mariam Elzahaby. A Hybrid Model for Stock Market Forecasting: Integrating News Sentiment and Time Series Data with Graph Neural Networks. *arXiv preprint arXiv:2512.08567*, 2025. URL <https://arxiv.org/abs/2512.08567>.
- [198] Matteo De Sanctis, Riccardo De Sanctis, Stefano Faralli, Paola Velardi, and Bardh Prenkaj. Beyond Edge Deletion: A Comprehensive Approach to Counterfactual Explanation in Graph Neural Networks. *arXiv preprint arXiv:2603.04209*, 2026. URL <https://arxiv.org/abs/2603.04209>.
- [199] Soumya Sanyal, Janakiraman Balachandran, Naganand Yadati, Abhishek Kumar, Padmini Rajagopalan, Suchismita Sanyal, and Partha Talukdar. MT-CGCNN: Integrating Crystal Graph Convolutional Neural Network with Multitask Learning for Material Property Prediction. *arXiv preprint arXiv:1811.05660*, 2018. URL <https://arxiv.org/abs/1811.05660>.
- [200] Rishov Sarkar, Stefan Abi-Karam, Yuqi He, Lakshmi Sathidevi, and Cong Hao. FlowGNN: A Dataflow Architecture for Real-Time Workload-Agnostic Graph Neural Network Inference. *arXiv preprint arXiv:2204.13103*, 2022. URL <https://arxiv.org/abs/2204.13103>.
- [201] Ruhi Sayana, Kate Callon, Jennifer Xu, Jonathan Deutsch, Steven Chu, James Zou, John Janetzko, Rabindra V. Shrivnaraine, and Kyle Swanson. Generating readily synthesizable small molecule fluorophore scaffolds with reinforcement learning. *arXiv preprint arXiv:2601.07145*, 2026. URL <https://arxiv.org/abs/2601.07145>.
- [202] Brandon Schoener, Yuting Hu, Pasit Wanlapha, Akshay Rengarajan, Ian Moog, Michael Wang, Peihong Zhang, Jinjun Xiong, and Hao Zeng. A New Workflow for Materials Discovery Bridging the Gap Between Experimental Databases and Graph Neural Networks. *arXiv preprint arXiv:2602.00756*, 2026. URL <https://arxiv.org/abs/2602.00756>.
- [203] Mathias Schreiner, Arghya Bhowmik, Tejs Vegge, Peter Bjørn Jørgensen, and Ole Winther. NeuralNEB – Neural Networks can find Reaction Paths Fast. *arXiv preprint arXiv:2207.09971*, 2022. URL <https://arxiv.org/abs/2207.09971>.
- [204] Artur M. Schweidtmann, Jan G. Rittig, Jana M. Weber, Martin Grohe, Manuel Dahmen, Kai Leonhard, and Alexander Mitsos. Physical Pooling Functions in Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2207.13779*, 2022. URL <https://arxiv.org/abs/2207.13779>.
- [205] Teng Jiek See, Daokun Zhang, Mario Boley, and David K. Chalmers. Layer-to-Layer Knowledge Mixing in Graph Neural Network for Chemical Property Prediction. *arXiv preprint arXiv:2510.20236*, 2025. URL <https://arxiv.org/abs/2510.20236>.
- [206] Aditya Sengar, Ali Hariri, Daniel Probst, Patrick Barth, and Pierre Vandergheynst. Generative Modeling of Full-Atom Protein Conformations using Latent Diffusion on Graph Embeddings. *arXiv preprint arXiv:2506.17064*, 2025. URL <https://arxiv.org/abs/2506.17064>.
- [207] Florian Sestak, Lisa Schneckenreiter, Johannes Brandstetter, Sepp Hochreiter, Andreas Mayr, and Gunter Klambauer. VN-EGNN: E(3)-Equivariant Graph Neural Networks with Virtual Nodes Enhance Protein Binding Site Identification. *arXiv preprint arXiv:2404.07194*, 2024. URL <https://arxiv.org/abs/2404.07194>.
- [208] Chen Shao, Zhou Chen, and Pascal Friederich. Graph neural networks to learn joint representations of disjoint molecular graphs. *arXiv preprint arXiv:2210.09517*, 2022. URL <https://arxiv.org/abs/2210.09517>.
- [209] Kieren Sharma, Lucia Marucci, and Zahraa S. Abdallah. FluxGAT: Integrating Flux Sampling with Graph Neural Networks for Unbiased Gene Essentiality Classification. *arXiv preprint arXiv:2403.18666*, 2024. URL <https://arxiv.org/abs/2403.18666>.
- [210] Mahmuda Sharmin, Taihao Han, Jie Huang, Narayanan Neithalath, Gaurav Sant, and Aditya Kumar. A Roadmap for Applying Graph Neural Networks to Numerical Data: Insights from Cementitious Materials. *arXiv preprint arXiv:2512.14855*, 2025. URL <https://arxiv.org/abs/2512.14855>.
- [211] Ji Shi, Chengxun Xie, Zhonghao Li, Xinming Zhang, and Miao Zhang. JTreeformer: Graph-Transformer via Latent-Diffusion Model for Molecular Generation. *arXiv preprint arXiv:2504.20770*, 2025. URL <https://arxiv.org/abs/2504.20770>.
- [212] Yaorui Shi, An Zhang, Enzhi Zhang, Zhiyuan Liu, and Xiang Wang. ReLM: Leveraging Language Models for Enhanced Chemical Reaction Prediction. *arXiv preprint arXiv:2310.13590*, 2023. URL <https://arxiv.org/abs/2310.13590>.
- [213] Zanyu Shi, Yang Wang, Pathum M. Weerawarna, Jie Zhang, Timothy I. Richardson, Yijie Wang, and Kun Huang. Structure-Aware Compound-Protein Affinity Prediction via Graph Neural Networks with Group Lasso Regularization. *arXiv preprint arXiv:2507.03318*, 2025. URL <https://arxiv.org/abs/2507.03318>.

- [214] Hiroyuki Shindo and Yuji Matsumoto. Gated Graph Recursive Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:1909.00259*, 2019. URL <https://arxiv.org/abs/1909.00259>.
- [215] Tomoya Shiota, Kenji Ishihara, and Wataru Mizukami. Universal neural network potentials as descriptors: Towards scalable chemical property prediction using quantum and classical computers. *arXiv preprint arXiv:2402.18433*, 2024. URL <https://arxiv.org/abs/2402.18433>.
- [216] Zeren Shui and George Karypis. Heterogeneous Molecular Graph Neural Networks for Predicting Molecule Properties. *arXiv preprint arXiv:2009.12710*, 2020. URL <https://arxiv.org/abs/2009.12710>.
- [217] Aman Singh, Shahid Shafi Dar, Ranveer Singh, and Nagendra Kumar. A Hybrid Similarity-Aware Graph Neural Network with Transformer for Node Classification. *arXiv preprint arXiv:2504.02615*, 2025. URL <https://arxiv.org/abs/2504.02615>.
- [218] Aryan Singh, Pepijn Van de Ven, Ciaran Eising, and Patrick Denny. Compact & Capable: Harnessing Graph Neural Networks and Edge Convolution for Medical Image Classification. *arXiv preprint arXiv:2307.12790*, 2023. URL <https://arxiv.org/abs/2307.12790>.
- [219] Anuj Kumar Sirohi, Subhanu Halder, Kabir Kumar, and Sandeep Kumar. Enhancing Robustness of Graph Neural Networks through p-Laplacian. *arXiv preprint arXiv:2409.19096*, 2024. URL <https://arxiv.org/abs/2409.19096>.
- [220] Anuj Kumar Sirohi, Subhanu Halder, Kabir Kumar, and Sandeep Kumar. Enhancing Robustness of Graph Neural Networks through p-Laplacian. *arXiv preprint arXiv:2511.06143*, 2025. URL <https://arxiv.org/abs/2511.06143>.
- [221] Zhilong Song, Linfeng Fan, Shuaihua Lu, Qionghua Zhou, Chongyi Ling, and Jinlan Wang. Inverse Design of Promising Alloys for Electrocatalytic CO₂ Reduction via Generative Graph Neural Networks Combined with Bird Swarm Algorithm. *arXiv preprint arXiv:2405.18891*, 2024. URL <https://arxiv.org/abs/2405.18891>.
- [222] Henry W. Sprueill, Carl Edwards, Khushbu Agarwal, Mariefel V. Olarte, Udishnu Sanyal, Conrad Johnston, Hongbin Liu, Heng Ji, and Sutanay Choudhury. ChemReasoner: Heuristic Search over a Large Language Model’s Knowledge Space using Quantum-Chemical Feedback. *arXiv preprint arXiv:2402.10980*, 2024. URL <https://arxiv.org/abs/2402.10980>.
- [223] Sakhinana Sagar Srinivas and Venkataramana Runkana. Cross-Modal Learning for Chemistry Property Prediction: Large Language Models Meet Graph Machine Learning. *arXiv preprint arXiv:2408.14964*, 2024. URL <https://arxiv.org/abs/2408.14964>.
- [224] Wojciech G. Stark, Cas van der Oord, Ilyes Batatia, Yaolong Zhang, Bin Jiang, Gabor Csanyi, and Reinhard J. Maurer. Benchmarking of machine learning interatomic potentials for reactive hydrogen dynamics at metal surfaces. *arXiv preprint arXiv:2403.15334*, 2024. URL <https://arxiv.org/abs/2403.15334>.
- [225] Bing Su, Dazhao Du, Zhao Yang, Yujie Zhou, Jiangmeng Li, Anyi Rao, Hao Sun, Zhiwu Lu, and Ji-Rong Wen. A Molecular Multimodal Foundation Model Associating Molecule Graphs with Natural Language. *arXiv preprint arXiv:2209.05481*, 2022. URL <https://arxiv.org/abs/2209.05481>.
- [226] YunHe Su, Zhengyang Lu, Junhui Liu, Ke Pang, Haoran Dai, Sa Liu, Yuxin Jia, Lujia Ge, and Jing min Yang. Applications of Large Models in Medicine. *arXiv preprint arXiv:2502.17132*, 2025. URL <https://arxiv.org/abs/2502.17132>.
- [227] Jose Suarez-Varela, Paul Almasan, Miquel Ferriol-Galmes, Krzysztof Rusek, Fabien Geyer, Xiangli Cheng, Xiang Shi, Shihan Xiao, Franco Scarselli, Albert Cabellos-Aparicio, and Pere Barlet-Ros. Graph Neural Networks for Communication Networks: Context, Use Cases and Opportunities. *arXiv preprint arXiv:2112.14792*, 2021. URL <https://arxiv.org/abs/2112.14792>.
- [228] Mengying Sun, Jing Xing, Huijun Wang, Bin Chen, and Jiayu Zhou. MoCL: Data-driven Molecular Fingerprint via Knowledge-aware Contrastive Learning from Molecular Graph. *arXiv preprint arXiv:2106.04509*, 2021. URL <https://arxiv.org/abs/2106.04509>.
- [229] Peng-Fei Sun, Chuan-Xian Ren, and Hong Yan. Curvature-Informed Potential Energy Surface for Protein-Ligand Binding Affinity Prediction. *arXiv preprint arXiv:2606.14217*, 2026. URL <https://arxiv.org/abs/2606.14217>.
- [230] Ruoxi Sun, Hanjun Dai, Li Li, Steven Kearnes, and Bo Dai. Energy-based View of Retrosynthesis. *arXiv preprint arXiv:2007.13437*, 2020. URL <https://arxiv.org/abs/2007.13437>.
- [231] Yundong Sun, Dongjie Zhu, Yansong Wang, Zhaoshuo Tian, Ning Cao, and Gregory O’Hared. SpikeGraphormer: A High-Performance Graph Transformer with Spiking Graph Attention. *arXiv preprint arXiv:2403.15480*, 2024. URL <https://arxiv.org/abs/2403.15480>.
- [232] Maciej Sypetkowski, Frederik Wenkel, Farimah Poursafaei, Nia Dickson, Karush Suri, Philip Fradkin, and Dominique Beaini. On the Scalability of GNNs for Molecular Graphs. *arXiv preprint*

- arXiv:2404.11568*, 2024. URL <https://arxiv.org/abs/2404.11568>.
- [233] Liqin Tan, Pin Chen, Menghan Liu, Xiean Wang, Jianhuan Cen, and Qingsong Zou. Benchmarking GNNs for OOD Materials Property Prediction with Uncertainty Quantification. *arXiv preprint arXiv:2511.11697*, 2025. URL <https://arxiv.org/abs/2511.11697>.
- [234] Han Tang, Shikun Feng, Bicheng Lin, Yuyan Ni, Jingjing Liu, Wei-Ying Ma, and Yanyan Lan. Contextual Molecule Representation Learning from Chemical Reaction Knowledge. *arXiv preprint arXiv:2402.13779*, 2024. URL <https://arxiv.org/abs/2402.13779>.
- [235] Shuo Tao and Li Zhu. EOSnet: Embedded Overlap Structures for Graph Neural Networks in Predicting Material Properties. *arXiv preprint arXiv:2411.02579*, 2024. URL <https://arxiv.org/abs/2411.02579>.
- [236] Kartar Kumar Lohana Tharwani, Rajesh Kumar, Sumita, Numan Ahmed, and Yong Tang. Large Language Models Transform Organic Synthesis From Reaction Prediction to Automation. *arXiv preprint arXiv:2508.05427*, 2025. URL <https://arxiv.org/abs/2508.05427>.
- [237] Luke Thompson, Davy Guan, Dai Shi, Slade Matthews, Junbin Gao, and Andi Han. ATOM: A Pretrained Neural Operator for Multitask Molecular Dynamics. *arXiv preprint arXiv:2510.05482*, 2025. URL <https://arxiv.org/abs/2510.05482>.
- [238] Hirofumi Tsuruta, Yukari Katsura, and Masaya Kumagai. DeepCrysTet: A Deep Learning Approach Using Tetrahedral Mesh for Predicting Properties of Crystalline Materials. *arXiv preprint arXiv:2310.06852*, 2023. URL <https://arxiv.org/abs/2310.06852>.
- [239] Zhengkai Tu and Connor W. Coley. Permutation invariant graph-to-sequence model for template-free retrosynthesis and reaction prediction. *arXiv preprint arXiv:2110.09681*, 2021. URL <https://arxiv.org/abs/2110.09681>.
- [240] Ryoichi Uchiyama, Yuya Nakajima, Yuta Tanaka, and Junji Seino. Impact of Local Descriptors Derived from Machine Learning Potentials in Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2602.03046*, 2026. URL <https://arxiv.org/abs/2602.03046>.
- [241] Disha Varshney, Samarth Garg, Sarthak Tyagi, Deeksha Varshney, Nayan Deep, and Asif Ekbal. Protein Secondary Structure Prediction Using 3D Graphs and Relation-Aware Message Passing Transformers. *arXiv preprint arXiv:2511.13685*, 2025. URL <https://arxiv.org/abs/2511.13685>.
- [242] Petar Velickovic. Everything is Connected: Graph Neural Networks. *arXiv preprint arXiv:2301.08210*, 2023. URL <https://arxiv.org/abs/2301.08210>.
- [243] Tirtha Vinchurkar, Kareem Abdelmaqsoud, and John R. Kitchin. Uncertainty Quantification in Graph Neural Networks with Shallow Ensembles. *arXiv preprint arXiv:2504.12627*, 2025. URL <https://arxiv.org/abs/2504.12627>.
- [244] Michael Vitz, Hamed Mohammadbagherpoor, Samarth Sandeep, Andrew Vlasic, Richard Padbury, and Anh Pham. Hybrid Quantum Graph Neural Network for Molecular Property Prediction. *arXiv preprint arXiv:2405.05205*, 2024. URL <https://arxiv.org/abs/2405.05205>.
- [245] Amanda A. Volk, Robert W. Epps, Jeffrey G. Ethier, and Luke A. Baldwin. Modeling Multi-Step Scientific Processes with Graph Transformer Networks. *arXiv preprint arXiv:2408.05425*, 2024. URL <https://arxiv.org/abs/2408.05425>.
- [246] Nikita Volzhin and Soowhan Yoon. Kolmogorov-Arnold graph neural networks for chemically informed prediction tasks on inorganic nanomaterials. *arXiv preprint arXiv:2512.19494*, 2025. URL <https://arxiv.org/abs/2512.19494>.
- [247] Brook Wander, Muhammed Shuaibi, John R. Kitchin, Zachary W. Ulissi, and C. Lawrence Zitnick. CatTSunami: Accelerating Transition State Energy Calculations with Pre-trained Graph Neural Networks. *arXiv preprint arXiv:2405.02078*, 2024. URL <https://arxiv.org/abs/2405.02078>.
- [248] J. Wang, X. Liu, S. Shen, L. Deng, and H. Liu*. DeepDDS: deep graph neural network with attention mechanism to predict synergistic drug combinations. *arXiv preprint arXiv:2107.02467*, 2021. URL <https://arxiv.org/abs/2107.02467>.
- [249] Xiyuan Wang and Muhan Zhang. Graph Neural Network with Local Frame for Molecular Potential Energy Surface. *arXiv preprint arXiv:2208.00716*, 2022. URL <https://arxiv.org/abs/2208.00716>.
- [250] Xu Wang, Huan Zhao, Lanning Wei, and Quanming Yao. Graph Property Prediction on Open Graph Benchmark: A Winning Solution by Graph Neural Architecture Search. *arXiv preprint arXiv:2207.06027*, 2022. URL <https://arxiv.org/abs/2207.06027>.
- [251] Yifei Wang, Yunrui Li, Lin Liu, Pengyu Hong, and Hao Xu. Advancing Drug Discovery with Enhanced Chemical Understanding via Asymmetric Contrastive Multimodal Learning. *arXiv preprint arXiv:2311.06456*, 2023. URL <https://arxiv.org/abs/2311.06456>.
- [252] Yinkai Wang, Yan Zhou Chen, Xiaohui Chen, Li-Ping Liu, and Soha Hassoun. SpecBridge: Bridging Mass Spectrometry and Molecular Representations via Cross-Modal Alignment. *arXiv preprint*

- arXiv:2601.17204*, 2026. URL <https://arxiv.org/abs/2601.17204>.
- [253] Yu Wang, Chao Pang, Yuzhe Wang, Yi Jiang, Junru Jin, Sirui Liang, Quan Zou, and Leyi Wei. MechRetro is a chemical-mechanism-driven graph learning framework for interpretable retrosynthesis prediction and pathway planning. *arXiv preprint arXiv:2210.02630*, 2022. URL <https://arxiv.org/abs/2210.02630>.
- [254] Yu Wang, JingJie Zhang, Junru Jin, and Leyi Wei. MolCAP: Molecular Chemical reActivity pretraining and prompted-finetuning enhanced molecular representation learning. *arXiv preprint arXiv:2306.09187*, 2023. URL <https://arxiv.org/abs/2306.09187>.
- [255] Yuelin Wang, Kai Yi, Xinliang Liu, Yu Guang Wang, and Shi Jin. ACMP: Allen-Cahn Message Passing with Attractive and Repulsive Forces for Graph Neural Networks. *arXiv preprint arXiv:2206.05437*, 2022. URL <https://arxiv.org/abs/2206.05437>.
- [256] Yusong Wang, Shaoning Li, Xinheng He, Mingyu Li, Zun Wang, Nanning Zheng, Bin Shao, Tie-Yan Liu, and Tong Wang. ViSNet: an equivariant geometry-enhanced graph neural network with vector-scalar interactive message passing for molecules. *arXiv preprint arXiv:2210.16518*, 2022. URL <https://arxiv.org/abs/2210.16518>.
- [257] Yusong Wang, Shaoning Li, Zun Wang, Xinheng He, Bin Shao, Tie-Yan Liu, and Tong Wang. An ensemble of VisNet, Transformer-M, and pretraining models for molecular property prediction in OGB Large-Scale Challenge @ NeurIPS 2022. *arXiv preprint arXiv:2211.12791*, 2022. URL <https://arxiv.org/abs/2211.12791>.
- [258] Yuyang Wang, Jianren Wang, Zhonglin Cao, and Amir Barati Farimani. Molecular Contrastive Learning of Representations via Graph Neural Networks. *arXiv preprint arXiv:2102.10056*, 2021. URL <https://arxiv.org/abs/2102.10056>.
- [259] Yuyang Wang, Zijie Li, and Amir Barati Farimani. Graph Neural Networks for Molecules. *arXiv preprint arXiv:2209.05582*, 2022. URL <https://arxiv.org/abs/2209.05582>.
- [260] Zhen Wang, Zheng Feng, Yanjun Li, Bowen Li, Yongrui Wang, Chulin Sha, Min He, and Xiaolin Li. BatmanNet: Bi-branch Masked Graph Transformer Autoencoder for Molecular Representation. *arXiv preprint arXiv:2211.13979*, 2022. URL <https://arxiv.org/abs/2211.13979>.
- [261] Zun Wang, Chong Wang, Sibao Zhao, Yong Xu, Shaogang Hao, Chang Yu Hsieh, Bing-Lin Gu, and Wenhui Duan. Heterogeneous relational message passing networks for molecular dynamics simulations. *arXiv preprint arXiv:2109.00711*, 2021. URL <https://arxiv.org/abs/2109.00711>.
- [262] Jason Wu and Petar Velickovic. Leveraging Classical Algorithms for Graph Neural Networks. *arXiv preprint arXiv:2510.21574*, 2025. URL <https://arxiv.org/abs/2510.21574>.
- [263] Kun Wu, Mert Hidayetoglu, Xiang Song, Sitao Huang, Da Zheng, Israt Nisa, and Wen mei Hwu. Hector: An Efficient Programming and Compilation Framework for Implementing Relational Graph Neural Networks in GPU Architectures. *arXiv preprint arXiv:2301.06284*, 2023. URL <https://arxiv.org/abs/2301.06284>.
- [264] Liming Wu, Rui Jiao, Qi Li, Mingze Li, Songyou Li, Shifeng Jin, and Wenbing Huang. DMFlow: Disordered Materials Generation by Flow Matching. *arXiv preprint arXiv:2602.04734*, 2026. URL <https://arxiv.org/abs/2602.04734>.
- [265] Shiguang Wu, Yaqing Wang, and Quanming Yao. PACIA: Parameter-Efficient Adapter for Few-Shot Molecular Property Prediction. *arXiv preprint arXiv:2310.00614*, 2023. URL <https://arxiv.org/abs/2310.00614>.
- [266] Yuran Xiang, Haiteng Zhao, Chang Ma, and Zhi-Hong Deng. Instruction-Based Molecular Graph Generation with Unified Text-Graph Diffusion Model. *arXiv preprint arXiv:2408.09896*, 2024. URL <https://arxiv.org/abs/2408.09896>.
- [267] Shufang Xie, Rui Yan, Peng Han, Yingce Xia, Lijun Wu, Chenjuan Guo, Bin Yang, and Tao Qin. RetroGraph: Retrosynthetic Planning with Graph Search. *arXiv preprint arXiv:2206.11477*, 2022. URL <https://arxiv.org/abs/2206.11477>.
- [268] Yutong Xie, Chence Shi, Hao Zhou, Yuwei Yang, Weinan Zhang, Yong Yu, and Lei Li. MARS: Markov Molecular Sampling for Multi-objective Drug Discovery. *arXiv preprint arXiv:2103.10432*, 2021. URL <https://arxiv.org/abs/2103.10432>.
- [269] Hongsheng Xing and Qiuxin Si. Learning Continuous Solvent Effects from Transient Flow Data: A Graph Neural Network Benchmark on Catechol Rearrangement. *arXiv preprint arXiv:2512.19530*, 2025. URL <https://arxiv.org/abs/2512.19530>.
- [270] Fanglei Xue, Meihan Zhang, Shuqi Li, Xinyu Gao, James A. Wohlschlegel, Wenbing Huang, Yi Yang, and Weixian Deng. SE(3)-Equivariant Ternary Complex Prediction Towards Target Protein Degradation. *arXiv preprint arXiv:2502.18875*, 2025. URL <https://arxiv.org/abs/2502.18875>.
- [271] Chaoqi Yang, Cao Xiao, Fenglong Ma, Lucas Glass, and Jimeng Sun. SafeDrug: Dual Molecular Graph Encoders for Recommending Effective and Safe Drug Combinations. *arXiv preprint*

- arXiv:2105.02711*, 2021. URL <https://arxiv.org/abs/2105.02711>.
- [272] Sherry Yang, Simon Batzner, Ruiqi Gao, Murathan Aykol, Alexander L. Gaunt, Brendan McMorrow, Danilo J. Rezende, Dale Schuurmans, Igor Mordatch, and Ekin D. Cubuk. Generative Hierarchical Materials Search. *arXiv preprint arXiv:2409.06762*, 2024. URL <https://arxiv.org/abs/2409.06762>.
- [273] Zhan’ao Yao, Boxuan Zhang, Jingyuan Shu, Xiaoyu Wu, Rongyan Wang, Linjing Li, Dajun Zeng, Yudong Yao, Tingwei Chen, Youwei Wang, Xiaolin Zhao, Jiahui Shi, and Jianjun Liu. MatMind: A Structure-Activity Knowledge-Driven Generative Foundation Model for Materials Science. *arXiv preprint arXiv:2606.07712*, 2026. URL <https://arxiv.org/abs/2606.07712>.
- [274] Shuqian Ye, Jiechun Liang, Rulin Liu, and Xi Zhu. Symmetrical Graph Neural Network for Quantum Chemistry, with Dual R/K Space. *arXiv preprint arXiv:1912.07256*, 2019. URL <https://arxiv.org/abs/1912.07256>.
- [275] Yingfang Yuan, Wenjun Wang, and Wei Pang. A Systematic Comparison Study on Hyperparameter Optimisation of Graph Neural Networks for Molecular Property Prediction. *arXiv preprint arXiv:2102.04283*, 2021. URL <https://arxiv.org/abs/2102.04283>.
- [276] Cheng Zeng, Harry W. Sullivan, Thomas Egg, Maya M. Martirosyan, Philipp Hollmer, Jirui Jin, Richard G. Hennig, Adrian Roitberg, Stefano Martiniani, Ellad B. Tadmor, and Mingjie Liu. MolCrystalFlow: Molecular Crystal Structure Prediction via Flow Matching. *arXiv preprint arXiv:2602.16020*, 2026. URL <https://arxiv.org/abs/2602.16020>.
- [277] Kaipeng Zeng, Bo yang, Xin Zhao, Yu Zhang, Fan Nie, Xiaokang Yang, Yaohui Jin, and Yanyan Xu. UAlign: Pushing the Limit of Template-free Retrosynthesis Prediction with Unsupervised SMILES Alignment. *arXiv preprint arXiv:2404.00044*, 2024. URL <https://arxiv.org/abs/2404.00044>.
- [278] Duna Zhan, Dongliang Guo, Pengsheng Ji, and Sheng Li. Bridging the Fairness Divide: Achieving Group and Individual Fairness in Graph Neural Networks. *arXiv preprint arXiv:2404.17511*, 2024. URL <https://arxiv.org/abs/2404.17511>.
- [279] Chenmu Zhang and Boris I. Yakobson. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. *arXiv preprint arXiv:2606.29717*, 2026. URL <https://arxiv.org/abs/2606.29717>.
- [280] Jintu Zhang, Luigi Bonati, Enrico Trizio, Odin Zhang, Yu Kang, TingJun Hou, and Michele Parrinello. Descriptors-free Collective Variables From Geometric Graph Neural Networks. *arXiv preprint arXiv:2409.07339*, 2024. URL <https://arxiv.org/abs/2409.07339>.
- [281] Odin Zhang, Haitao Lin, Xujun Zhang, Xiaorui Wang, Zhenxing Wu, Qing Ye, Weibo Zhao, Jike Wang, Kejun Ying, Yu Kang, Chang yu Hsieh, and Tingjun Hou. Graph Neural Networks in Modern AI-aided Drug Discovery. *arXiv preprint arXiv:2506.06915*, 2025. URL <https://arxiv.org/abs/2506.06915>.
- [282] Rong Zhang, Rongqing Yuan, and Boxue Tian. PointGAT: A quantum chemical property prediction model integrating graph attention and 3D geometry. *arXiv preprint arXiv:2310.05217*, 2023. URL <https://arxiv.org/abs/2310.05217>.
- [283] Shanzhuo Zhang, Lihang Liu, Sheng Gao, Donglong He, Xiaomin Fang, Weibin Li, Zhengjie Huang, Weiyue Su, and Wenjin Wang. LiteGEM: Lite Geometry Enhanced Molecular Representation Learning for Quantum Property Prediction. *arXiv preprint arXiv:2106.14494*, 2021. URL <https://arxiv.org/abs/2106.14494>.
- [284] Shuo Zhang, Yang Liu, and Lei Xie. Molecular Mechanics-Driven Graph Neural Network with Multiplex Graph for Molecular Structures. *arXiv preprint arXiv:2011.07457*, 2020. URL <https://arxiv.org/abs/2011.07457>.
- [285] Shuo Zhang, Yang Liu, and Lei Xie. Efficient and Accurate Physics-aware Multiplex Graph Neural Networks for 3D Small Molecules and Macromolecule Complexes. *arXiv preprint arXiv:2206.02789*, 2022. URL <https://arxiv.org/abs/2206.02789>.
- [286] Xiang Zhang, Zichun Zhou, Chen Ming, and Yi-Yang Sun. GPT-assisted learning of structure-property relationships by graph neural networks: Application to rare-earth doped phosphors. *arXiv preprint arXiv:2306.14238*, 2023. URL <https://arxiv.org/abs/2306.14238>.
- [287] Xuan Zhang, Haiyang Yu, Chengdong Wang, Jacob Helwig, Shuiwang Ji, and Xiaofeng Qian. Orbital Transformers for Predicting Wavefunctions in Time-Dependent Density Functional Theory. *arXiv preprint arXiv:2603.03511*, 2026. URL <https://arxiv.org/abs/2603.03511>.
- [288] Yang Zhang, Zhewei Wei, Ye Yuan, Chongxuan Li, and Wenbing Huang. EquiPocket: an E(3)-Equivariant Geometric Graph Neural Network for Ligand Binding Site Prediction. *arXiv preprint arXiv:2302.12177*, 2023. URL <https://arxiv.org/abs/2302.12177>.
- [289] Yuelin Zhang, Jiacheng Cen, Jiaqi Han, and Wenbing Huang. Fast and Distributed Equivariant Graph Neural Networks by Virtual Node Learn-

- ing. *arXiv preprint arXiv:2506.19482*, 2025. URL <https://arxiv.org/abs/2506.19482>.
- [290] Tianze Zheng, Ailun Wang, Xu Han, Yu Xia, Xingyuan Xu, Jiawei Zhan, Yu Liu, Yang Chen, Zhi Wang, Xiaojie Wu, Sheng Gong, and Wen Yan. Data-Driven Parametrization of Molecular Mechanics Force Fields for Expansive Chemical Space Coverage. *arXiv preprint arXiv:2408.12817*, 2024. URL <https://arxiv.org/abs/2408.12817>.
- [291] Tongya Zheng, Tianli Zhang, Qingzheng Guan, Wenjie Huang, Zunlei Feng, Mingli Song, and Chun Chen. Graph Neural Networks-based Hybrid Framework For Predicting Particle Crushing Strength. *arXiv preprint arXiv:2307.13909*, 2023. URL <https://arxiv.org/abs/2307.13909>.
- [292] Zinan Zheng, Yang Liu, Jia Li, Jianhua Yao, and Yu Rong. Relaxing Continuous Constraints of Equivariant Graph Neural Networks for Physical Dynamics Learning. *arXiv preprint arXiv:2406.16295*, 2024. URL <https://arxiv.org/abs/2406.16295>.
- [293] Artem Zholus, Maksim Kuznetsov, Roman Schutski, Rim Shayakhmetov, Daniil Polykovskiy, Sarath Chandar, and Alex Zhavoronkov. BindGPT: A Scalable Framework for 3D Molecular Design via Language Modeling and Reinforcement Learning. *arXiv preprint arXiv:2406.03686*, 2024. URL <https://arxiv.org/abs/2406.03686>.
- [294] Haoyu Zhou, Ping Xue, Hao Zhang, and Tianfan Fu. Quantized SO(3)-Equivariant Graph Neural Networks for Efficient Molecular Property Prediction. *arXiv preprint arXiv:2601.02213*, 2026. URL <https://arxiv.org/abs/2601.02213>.
- [295] Jie Zhou, Ganqu Cui, Shengding Hu, Zhengyan Zhang, Cheng Yang, Zhiyuan Liu, Lifeng Wang, Changcheng Li, and Maosong Sun. Graph Neural Networks: A Review of Methods and Applications. *arXiv preprint arXiv:1812.08434*, 2018. URL <https://arxiv.org/abs/1812.08434>.
- [296] Hao Zhu, Liping Liu, and Soha Hassoun. Using Graph Neural Networks for Mass Spectrometry Prediction. *arXiv preprint arXiv:2010.04661*, 2020. URL <https://arxiv.org/abs/2010.04661>.
- [297] Jianshen Zhu, Naveed Ahmed Azam, Kazuya Haraguchi, Liang Zhao, and Tatsuya Akutsu. Combining Graph Neural Networks and Mixed Integer Linear Programming for Molecular Inference under the Two-Layered Model. *arXiv preprint arXiv:2507.03920*, 2025. URL <https://arxiv.org/abs/2507.03920>.
- [298] Pingfei Zhu, Hongyi Liu, Xueyan Liu, Zhenjun Yang, and Bo Yang. GenShin: Guiding Rational Liposome Design by Ranking Liposomal Protein Corona through a Docking-Pose-Free GNN. *arXiv preprint arXiv:2504.13853*, 2025. URL <https://arxiv.org/abs/2504.13853>.
- [299] Weimin Zhu, Yi Zhang, DuanCheng Zhao, Jianrong Xu, and Ling Wang. HiGNN: Hierarchical Informative Graph Neural Networks for Molecular Property Prediction Equipped with Feature-Wise Attention. *arXiv preprint arXiv:2208.13994*, 2022. URL <https://arxiv.org/abs/2208.13994>.
- [300] Yanqiao Zhu, Jeehyun Hwang, Keir Adams, Zhen Liu, Bozhao Nan, Brock Stenfors, Yuanqi Du, Jatin Chauhan, Olaf Wiest, Olexandr Isayev, Connor W. Coley, Yizhou Sun, and Wei Wang. Learning Over Molecular Conformer Ensembles: Datasets and Benchmarks. *arXiv preprint arXiv:2310.00115*, 2023. URL <https://arxiv.org/abs/2310.00115>.